

THERMAL TRANSFER OVERPRINTER

(ENG) English



Version: 27.01.25 (V7)



ELECTRONIC SERVICE MANUAL

SVM 20c Series 32 CK



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1

PRINTER SETTINGS

Printer settings provides to;

- Change some values (delay, contrast and print position) while printer was printing.
- Edit system date and time.
- Adjust parameters.
- Load template to system.
- Choose and edit template.
- Change system language.
- Control sensors, valves, motors and fuses.
- Change IP host number.
- Entering the desired print quantity.

1.1 MAIN MENU

When the machine started, the following screen will appear. The functions on the main menu are explained one by one.

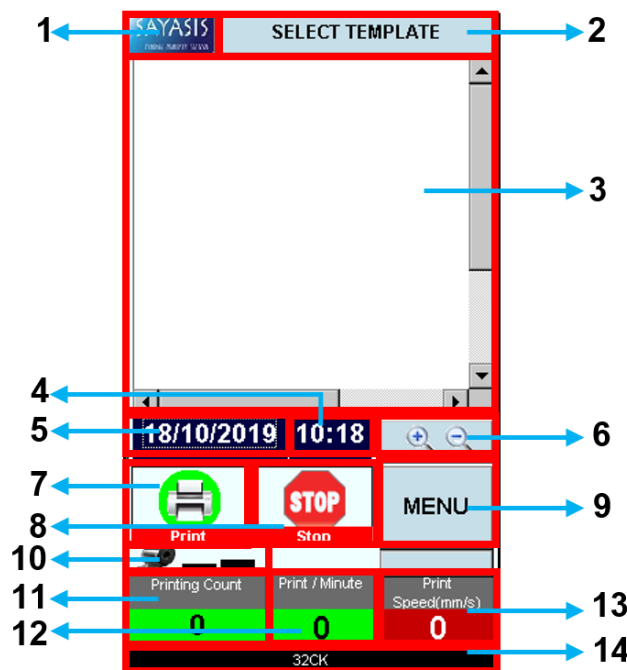


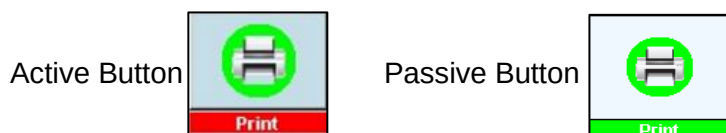
Figure 1. Main Menu

No	Designation
1	SAYASIS LOGO
2	SELECT TEMPLATE
3	TEMPLATE SCREEN
4	SYSTEM TIME
5	SYSTEM DATE
6	ZOOM
7	PRINT START
8	PRINT STOP
9	MENU
10	REMAINING RIBBON
11	PRINTING COUNT
12	PRINT/MINUTE
13	PRINT SPEED
14	MACHINE TYPE

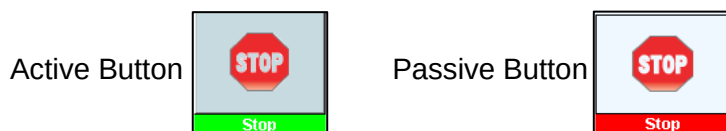
Table 1. Main Menu Functions

1. SAYASIS LOGO: This logo provides to open hidden settings window. When the clicked on this logo, system wants to enter password that is system date. Password format is "dmyyyy". Example 1; password is "18102019" for this date "18/10/2019". Example 2; password is "952019" for this date "09/05/2019" (zero at the beginning of the month and day is not entered in the password). After password is entered correctly, the hidden settings screen is opened. The screen provides to adjust parameter that variables to the general work of the machine. Also it provides to learn Information's of customer and machine ID. Authorization that variables of network adjusting is made from hidden settings screen. The logo is passive when printer started to print. If printer stops, the logo becomes active.

2. **SELECT TEMPLATE:** If the module doesn't have any template, the template must be installed from USB to internal memory. The Print and Stop functions will not be active until template has been selected. When the first template is loaded to system, Print function is active. If desired to change template after printer started, no need to stop the printer. This function (select template) is active every time (printer working state is on or off) after first working.
3. **TEMPLATE SCREEN:** This area shows chosen template. The template can't change via this screen. System will not response when was touched on this area. On this area, template can be scrolled to right or left via horizontal scroll bar and to up or down via vertical scroll bar. The scrolling does not affect printing.
4. **SYSTEM TIME:** This area shows the current system time. When the clicked to system time area, the time can be increased or decreased respectively "↑" and "↓" symbols. This function is active every time so the system time can be changed every time printer works or not.
5. **SYSTEM DATE:** This area shows the current system date. The correct password for hidden setting is this date. If today is 06/05/2019 but system date is 02/01/2008, the correct password is 212008 not 652019. When the clicked to system date area, the date can be increased or decreased respectively "↑" and "↓" symbols. This function is active every time so the system date can be changed every time printer works or not.
6. **ZOOM:** This function provides to zoom in or zoom out to selected template. Unless the template is selected, zoom feature is not active. Zooming in or out not affect template that printed. Only shows template from different scale.
7. **PRINT START:** This button is used to start print. There is a condition to activate this function. The template should be chosen. The active and passive displays of the button are shown below.



8. **PRINT STOP:** This button is used to stop print. There are 2 conditions to activate this function. First of all the template should be chosen and secondly printing should be started. The active and passive displays of the button are shown below.



9. **MENU:** This button opens the screen that have functions provide to load template to system, choose and edit template and control sensors, valves, motors and fuses. The printer must be stopped to control sensors, valves, motors and fuses. The other situation, the printer doesn't have to stop. The menu button is active every time. Buttons that appear blue in this menu are active.
10. **REMAINING RIBBON:** This field shows how much ribbon is remain. Line length decreases as the ribbon decreases. This field is for illustration only. No value can be entered.

11. **PRINTING COUNT:** Shows the number of prints that printed since the machine was started. This field is for illustration only. No value can be entered. The value is reset in the area when the machine is restarted. Even if the template is changed, this value not reset.
12. **PRINT/MINUTE:** Shows the number of prints that printed in one minute. This field is for illustration only. No value can be entered. The value is reset in the area when the machine is restarted. Even if the template is changed, this value not reset.
13. **PRINT SPEED:** Shows the speed of lines. The unit is mm/s. This field is for illustration only. No value can be entered. The value is reset in the area when the machine is stopped.
14. **MACHINE TYPE:** This area shows whether it is pneumatic or magnetic and also shows whether the intermittent or continuous. This field is for illustration only. No value can be entered.

1.1.1 SELECT TEMPLATE SECTION

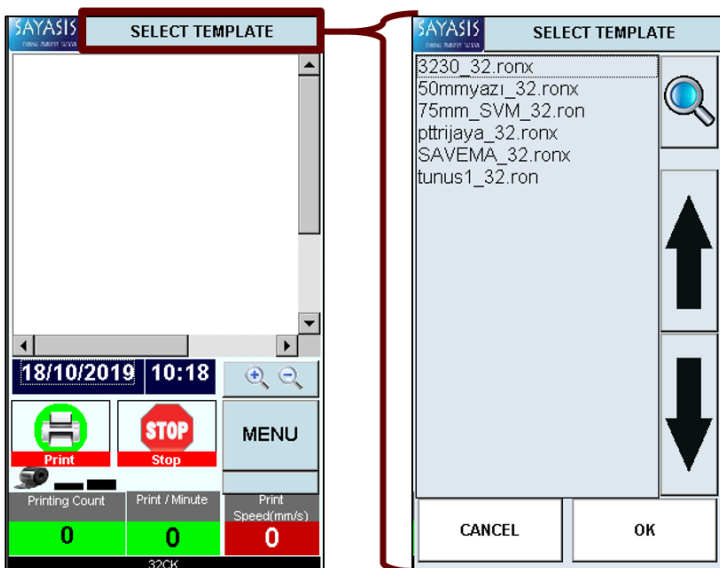


Figure 2. Template List Screen When Select Template Button Clicked

If the module doesn't have any template, the template must be installed from USB to internal memory. Choose desired template by using up or down arrow signs and press **OK** button. Unless press the button or if press **CANCEL** button, the chosen template is invalid. Press **OK** button before **CANCEL** button to be selected template is valid. After a five-second wait, selected template will appear on the screen.

1.1.2 SYSTEM DATE SECTION

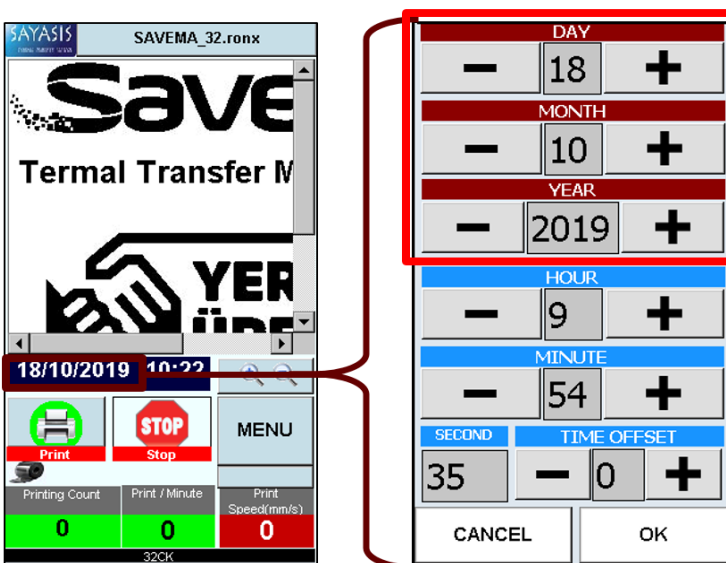


Figure 3. System Date Changing Screen When Date Button Clicked

System date can be changed with these settings. The date can be increased or decreased by using "↑" and "↓" buttons. System date is used for hidden settings and update templates. After the changes are completed, press **OK** button. Unless press the button or if press **CANCEL** button, the changes don't save. After pressing the **OK** button, if press **CANCEL** button, the changes are saved.

1.1.3 SYSTEM TIME SECTION

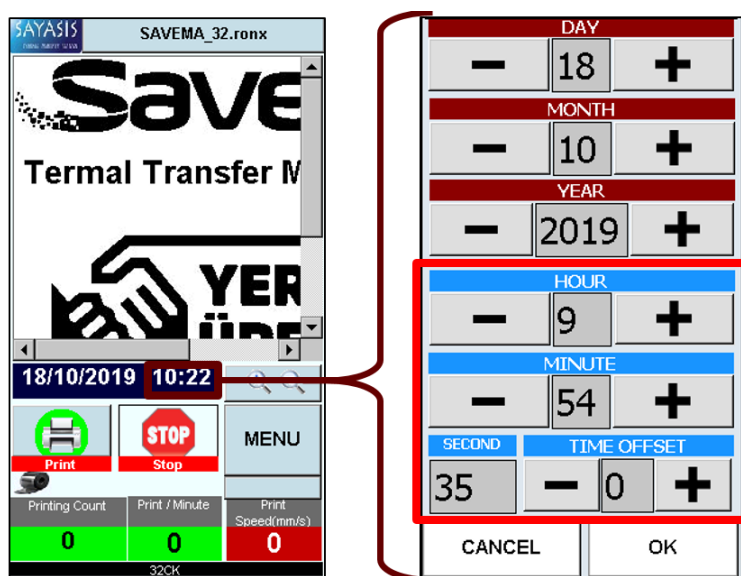


Figure 4. System Time Changing Screen When Time Button Clicked

System time can be changed with these settings. The date can be increased or decreased by using "↑" and "↓" buttons. System can be used time for update templates. After the changes are completed, press **OK** button. Unless press the button or if press **CANCEL** button, the changes don't save. After pressing the **OK** button, if press **CANCEL** button, the changes are saved.

1.1.4 MENU FUNCTIONS

Menu functions provide to load template to system, edit template and control sensors, valves, motors and fuses etc. Also some operations related to database and software settings can be done from this menu. The printer must be stopped to entering control section. The other situation, the printer doesn't have to stop. Buttons that appear blue are active.

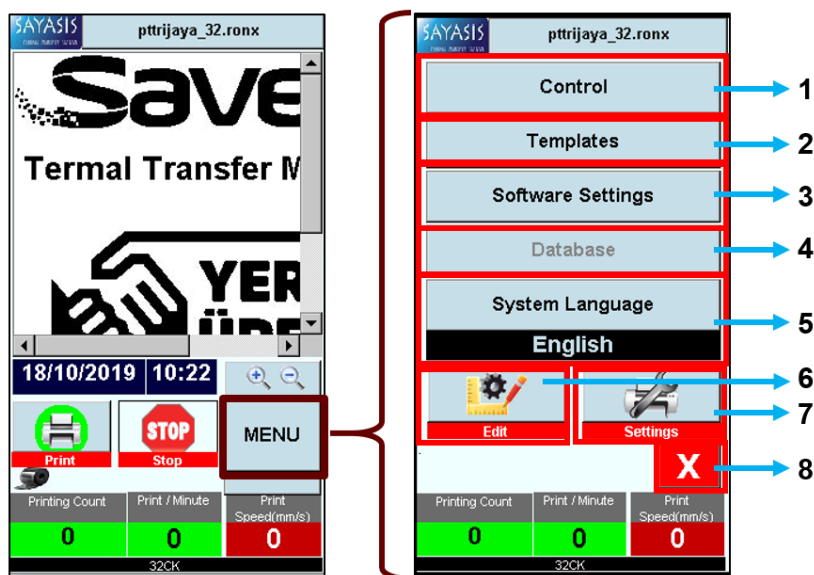


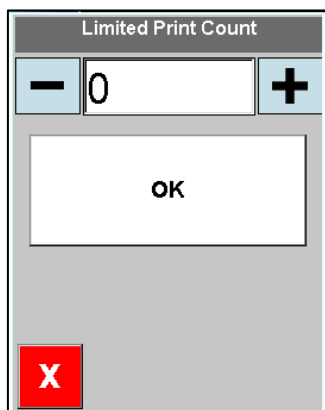
Figure 5. Menu Functions Screen

No	Designation
1	CONTROL
2	TEMPLATES
3	SOFTWARE SETTINGS
4	DATABASE
5	SYSTEM LANGUAGE
6	EDIT
7	SETTINGS
8	EXIT

Table 2. Menu Functions

1. **CONTROL:** Printer sensors, fuses, valves and motors can be controlled in this section. Also some signals that come to printer can be observed from the window. This function is not active while printer is printing. The printer must be stopped to enable this feature.
2. **TEMPLATES:** Templates can be saw in printer memory and USB stick memory. Templates can be inserted from USB stick to printer memory. The button is active every time.

3. **SOFTWARE SETTINGS:** Software related operations can be made. See product user manual for detailed information.
4. **DATABASE:** Database related operations can be made. See product user manual for detailed information.
5. **SYSTEM LANGUAGES:** System languages can be changed via this section. The area is active every time.
6. **EDIT:** Some components of the template (barcode, text, monetary value etc.) can be edited via this property. The template must be selected to enable this feature.
7. **SETTINGS:** Settings window provides to change some values (delay, contrast and print position etc.) while printer was printing. Also ribbonsave mode and mirroring settings can be done on this menu. Unless the template is selected, settings button is not active.
8. **EXIT:** It is used to go back main menu.

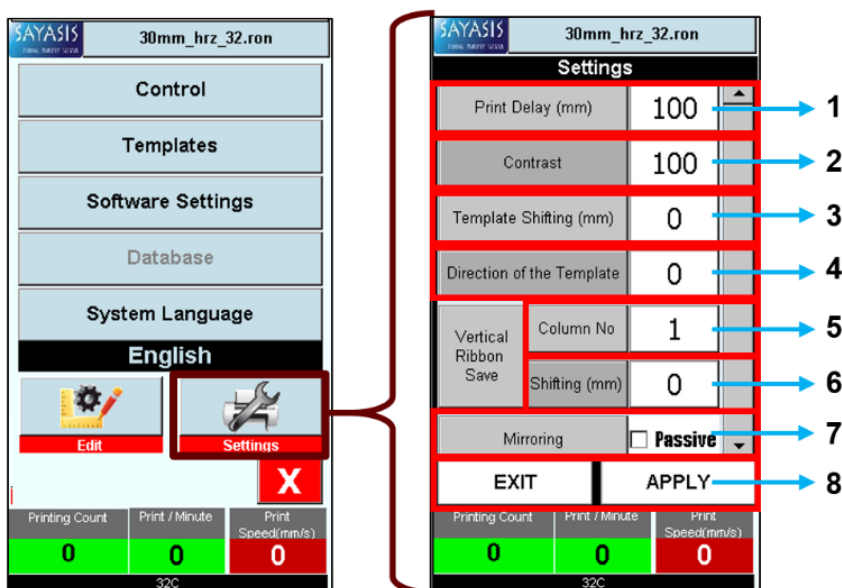


LIMITED PRINT COUNT: Value that desired template quantity can be entered with the help of the “+” and “-” buttons on drop-down area. When clicked “+” button increases value while “-” button is decreases. For example; if 5000 is entered in this area, printer counts back from 5000. If the value equals to 0, printer stops. The value is reset in the area when the machine is restarted. If the template is changed, this value does not reset. There is a condition to activate this function. This condition is that template should be chosen.

First press the menu button to access this field. The “print / minute”, “printing count” and “print / minute” fields are then pressed sequentially.

Figure 6. Screen when Template Counter Clicked

1.1.4.1 SETTINGS SECTION



Settings window provides to change some values (delay, contrast, template shifting and template direction) while printer was printing.

Ribbonsave mode and mirroring settings can be done at printing mode section on this menu. Unless the template is selected, settings button is not active.

Figure 7. Settings Screen When Settings Button Clicked

No	Designation
1	PRINT DELAY
2	CONTRAST
3	TEMPLATE SHIFTING
4	DIRECTION OF THE TEMPLATE
5	VERTICAL RIBBONSAVE/COLUMN NO
6	VERTICAL RIBBONSAVE/SHIFTING
7	MIRRORING
8	OPERATING BUTTONS

Table 3. Settings Functions

1. **PRINT DELAY:** Printer waits until the entered time in this box after contact signal came and does nothing. Printer prints after time-out. The unit is 10ms. That's mean; if 10 is entered to value, printer waits 100 ms. “+” button increases the delay value and print shifts more. “-” button decreases the delay value and print shifts less. The delay time cannot be negative. The maximum limit for delay time is 1000. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
2. **CONTRAST:** This function adjusts print darkness. The unit is none. “+” button increases the darkness value and print becomes darker. “-” button decreases the darkness value and print becomes lighter. The standard value is 100. The changeable range is 60-120 for contrast value. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
3. **TEMPLATE SHIFTING:** This function provides to determine the printing position along the thermal print head. The unit is mm. The shifting range is adjusted automatically by the printer according to the template. Positive values shift the template to the right and negative values shift the template to the left. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print. If the template in the picture below is shifted by 15mm, the template will look like the 2nd template below

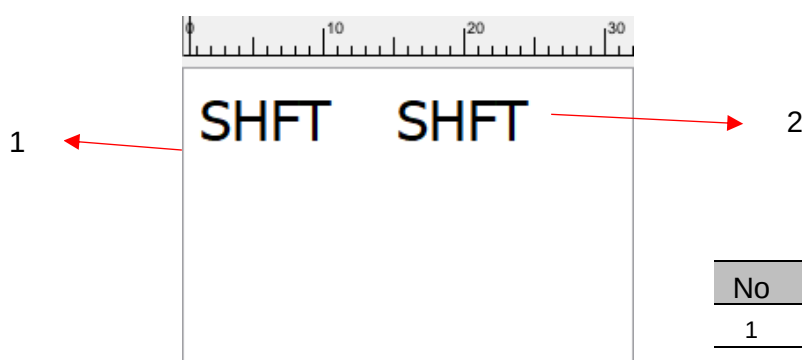


Figure 8 Template Shifting

No	Designation
1	MAIN TEMPLATE
2	SHIFTING TEMPLATE

Table 4 Template Shifting

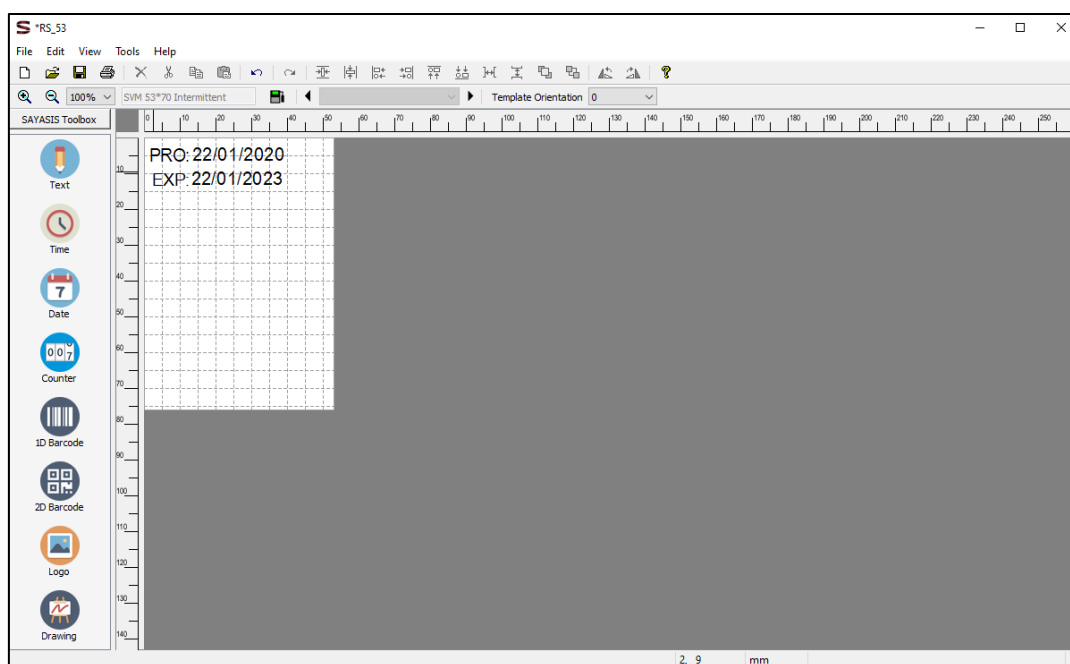
4. **DIRECTION OF THE TEMPLATE:** This feature provides to rotate template. By clicking down arrow sign, the desired rotation angle can be selected. Template can be rotated to right 0, 90, 180, 270 degree. The rotation angle should not considered by summing. Rotation process is performed by taking 0 point is a reference. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
5. **VERTICAL RIBBONSAVE/COLUMN NO:** This feature provides to more than one print in same column. Printer prints up the selected number. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
6. **VERTICAL RIBBONSAVE/SHIFTING:** This feature provide to shift between two print out in same row after ribbonsave mode/column no is entered more than one. The unit is mm. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
There are 2 options to use the vertical ribbonsave function.

Option 1;

Template must be prepared based on the top left corner and direction of template must be 90°.

Option 2;

Template must be prepared based on the upper right corner and direction of template must be 0°.



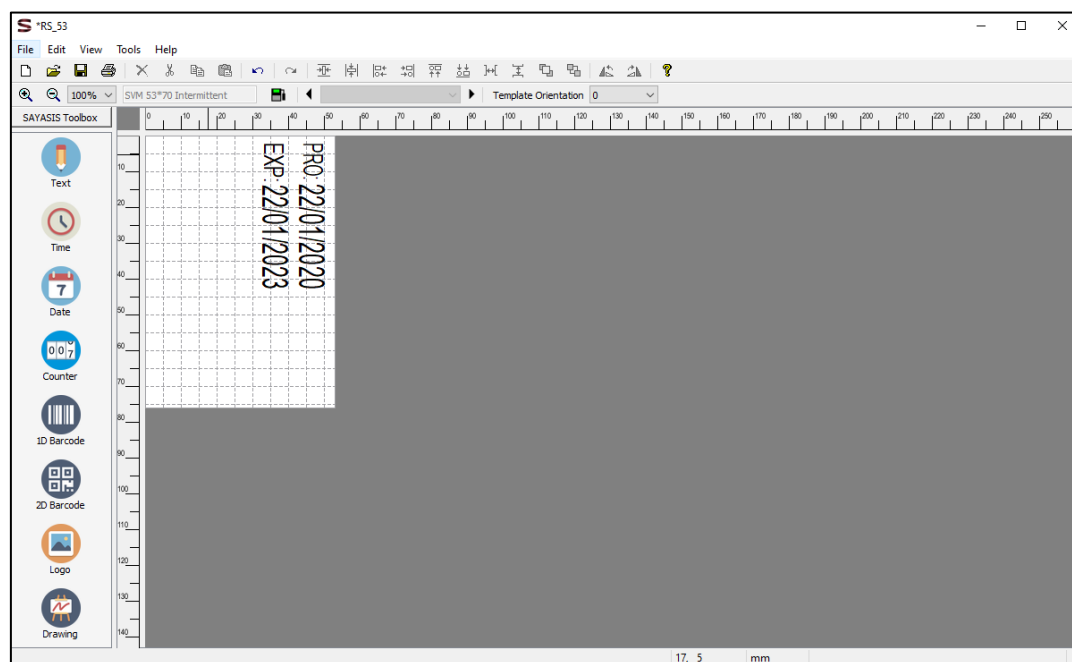


Figure 9. Preparation of Template Suitable for Vertical Ribbonsave Function - Option 1 (UP) and Option 2 (DOWN)

The following illustration (Figure 10) is drawn for VERTICAL RIBBONSAVE/COLUMN NO is 3. The distance between prints is automatically adjusted according to the value entered in VERTICAL RIBBONSAVE/SHIFTING. If ribbonsave function is used, the same column of ribbon is used for 3 prints since the VERTICAL RIBBONSAVE /COLUMN NO is entered 3. The point to note here is that if the function is used, printing on the package is not continuously done in the same place.

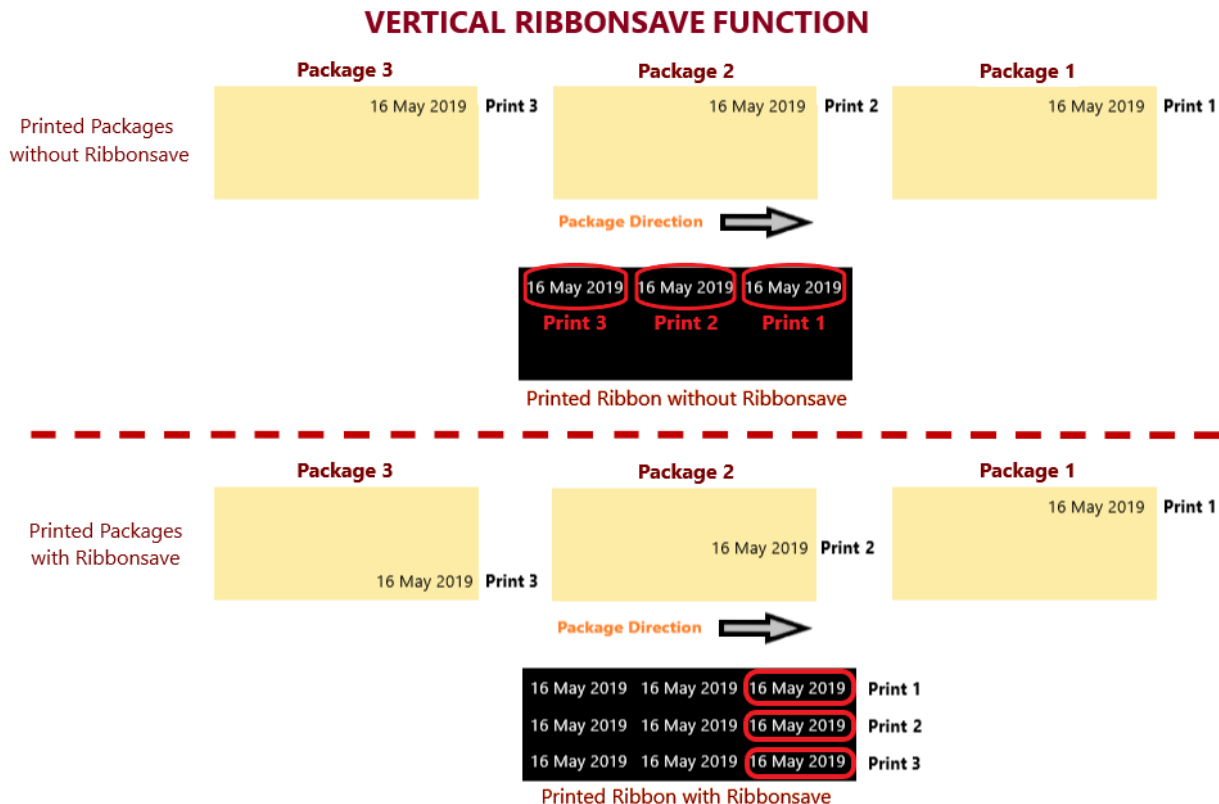


Figure 10. Ribbonsave Function

7. **MIRRORING**: This function provides to mirror template. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

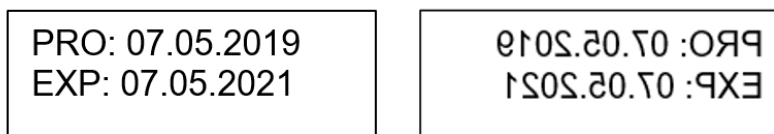


Figure 11. Original Template (Left) and Mirrored Template (Right)

8. **OPERATING BUTTONS**: The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

1.1.4.2 CONTROL – CONTROL SECTION

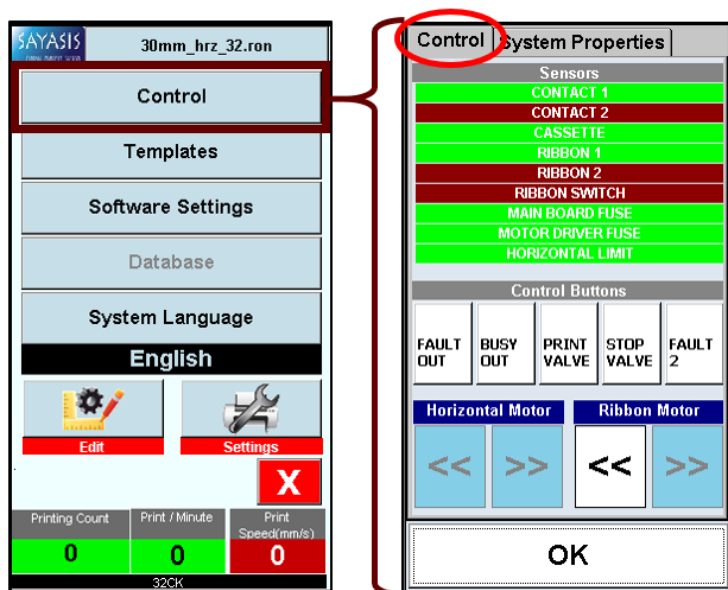


Figure 12. Control Screen When Control Button Clicked

Printer sensors, fuses, valves and motors can be controlled in this section. Also some signals that come to printer can be observed from the window. This function is not active while printer is printing. The printer must be stopped to enable this feature.

- **CONTACT 1**: Shows whether or not there is contact 1 signal. This field is for illustration only. System does not respond when clicked. If there is a contact signal, the field's color is green. Otherwise, color is red.
- **CONTACT 2**: Shows whether or not there is contact 2 signal. This field is for illustration only. System does not respond when clicked. If there is a contact signal, the field's color is green. Otherwise, color is red.
- **CASSETTE**: Shows whether or not there is cassette. This field is for illustration only. System does not respond when clicked. If the cassette is inserted, the field's color is green. Otherwise, color is red.
- **RIBBON 1**: This is used to check if the ribbon is broken. This field is for illustration only. System does not respond when clicked. If the ribbon is broken. The field's color is red. Otherwise, color is green.
- **RIBBON 2**: Shows that the ribbon is turning left or right. This field is for illustration only. System does not respond when clicked. If the ribbon is turning, the field's color is green. Otherwise, color is red.
- **RIBBON SWITCH**: Shows whether the ribbon break switch is working. When the switch is pressed, the color turns green.
- **MAINBOARD FUSE**: Shows whether the main board fuse is damaged. This field is for illustration only. System does not respond when clicked. If the fuse is damaged, the field's color is red. Otherwise, color is green.

- **MOTOR DRIVER FUSE:** Shows whether the main board fuse is damaged. This field is for illustration only. System does not respond when clicked. If the fuse is damaged, the field's color is red. Otherwise, color is green.
- **FAULT OUT:** This used to check the printer fault relay. The button is darker when clicked and relay connections that are normally open becomes normally close state.
- **BUSY OUT:** This used to check the printer busy relay. The button is darker when clicked and relay connections that are normally open becomes normally close state.
- **PRINT VALVE:** This used to test whether the print piston is working. The button is darker when clicked and thermal print head goes to print position. **ATTENTION: Finger should not be placed under the thermal head while this feature is used.**
- **STOP VALVE:** This used to test whether the stop piston is working. The button is darker when clicked and thermal print head goes as far as it goes. **ATTENTION: Fingers should not be placed under the thermal head while this feature is used.**
- **FAULT 2:** This used to check the printer fault mosfet. The button is darker when clicked and the circuit is completed through mosfet.(See 2.2.3 for more detail about fault mosfet connection)
- **RIBBON MOTOR:** This used to detect if the ribbon motor is turning left or right. The button is darker when right direction button was clicked and ribbon motor turns right. Also the button is darker when left direction button was clicked and ribbon motor turns left.
- **HORIZONTAL MOTOR:** This used to detect if the horizontal motor is moving left or right. The button is darker when right direction button was clicked and motor moves thermal print head to right. Also the button is darker when left direction button was clicked and motor moves thermal print head to left. **(This feature is not available because this printer does not have a horizontal motor.)**

1.1.4.3 CONTROL – INFORMATION SECTION

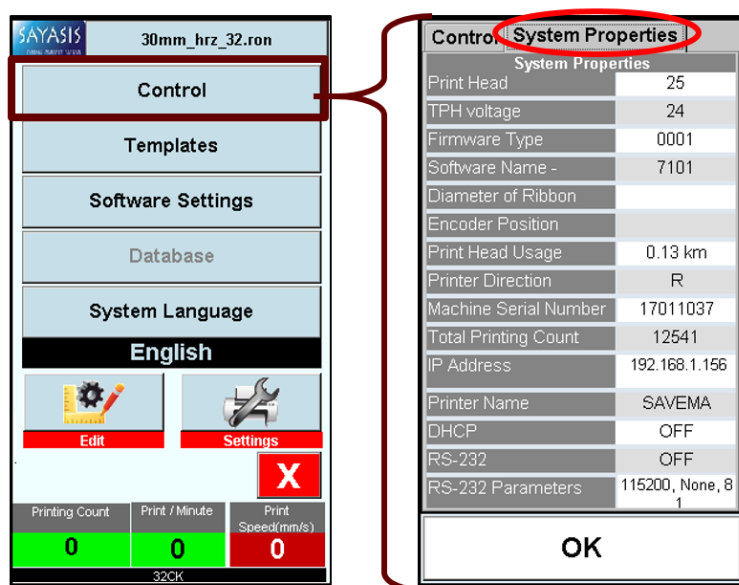


Figure 13. Information Screen When Control Button Clicked

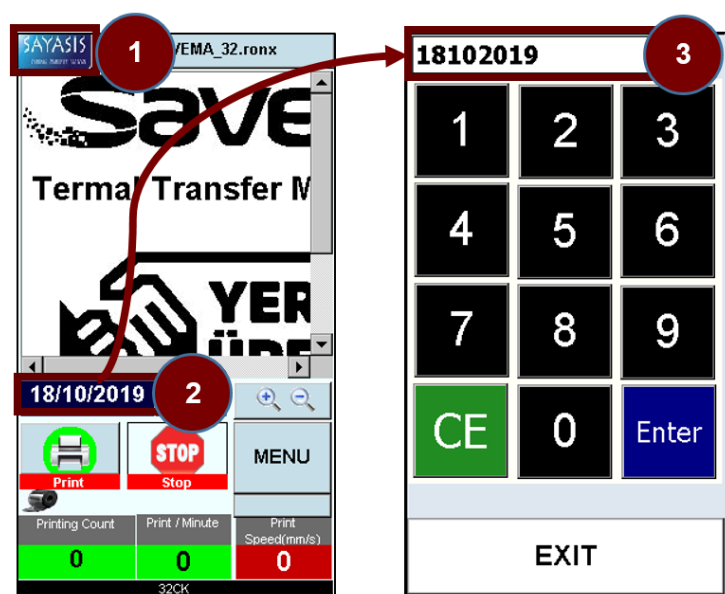
This field provides to take some information about machine. These fields are for illustration only. System does not respond when clicked.

- **PRINT HEAD TEMPERATURE (°C):** This field gives information about momentary print head temperature.
- **TPH VOLTAGE:** This field gives information about momentary print head voltage.
- **FIRMWARE TYPE:** This field gives information about machine firmware.
- **SOFTWARE NAME:** The name of the software installed on the system.
- **DIAMETER OF RIBBON:** This field gives information about used ribbon diameter.

- ENCODER POSITION: If the encoder is turning at correct direction, the value increases. If the encoder is turning at wrong direction, the value decreases.
- PRINT HEAD USAGE: This field gives information about amount of thermal head used in km from the first print. The value is never reset.
- PRINTER DIRECTION: This field gives information about printer direction (**Left/Right**).
- MACHINE SERIAL NUMBER: This field gives information about machine serial number.
- TOTAL PRINTING COUNT: Shows the number of prints from the first print. The value is never reset, neither the machine is reset nor the template is changed.
- IP ADDRESS: This field gives information about machine IP address.
- PRINTER NAME: This field gives information about name of machine.
- DHCP: This field gives information about automatic IP address acquisition (**ON/OFF**).
- RS-232: This field gives information about connection of RS-232 (**ON/OFF**).
- RS-232 PARAMETER: This field gives information about RS-232 parameters.

OK BUTTON: Provides to return to the menu function screen.

1.1.5 HIDDEN SETTINGS



No	Designation
1	SAYASIS LOGO
2	SYSTEM DATE
3	PASSWORD AREA

Table 5. Hidden Settings Signs

When the clicked on SAYASIS logo at main menu [1], system wants to enter password that is system date. Password format is "dmyyyy". Example 1; password is "18102019" for this date "18/10/2019". Example 2; password is "952019" for this date "09/05/2019". After password is entered correctly (after entering the value in [2] into [3]), the hidden settings screen is opened. The screen provides to adjust parameter that variables to the general work of the machine.

Figure 14. Password Screen When SAYASIS Button Clicked

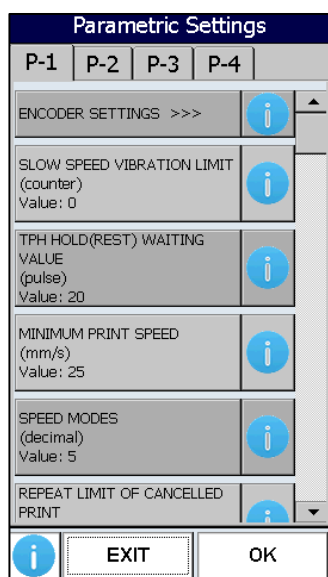
1.1.5.1 PARAMETRIC SETTINGS

Figure 15. Parameter Menu

There are variables to the general work of the machine in this section. These settings provide the opportunity to make improvements in printing with relevant parameters.

VALUE CHANGE SYSTEM:

1. Select parameter by scrolling through pages and change the value by numeric numbers.
2. Push the **"APPLY"** button and **"OK"** button at right bottom corner sequentially.
3. Push **"OK"** button and **"EXIT"** button at right bottom corner sequentially.

Note: If **"EXIT"** button is clicked before changes haven't been saved, there is no change in parameters.

Page No	Parameter Name	Sub-parameter Name	Minimum	Default	Maximum	Unit
1	Encoder Settings	1) Encoder Wheel Diameter	40	48	60	mm
		2) Encoder Pulse Number	500	1000	2500	pulse
		3) Encoder Reading Period	1	2	8	ms
		4) Encoder Rotation Direction	0	1	1	decimal
	5) Slow Speed Vibration Limit		0	20	200	counter
	6) Tph Hold(Rest) Waiting Value		0	20	300	pulse
	7) Minimum Print Speed		0	25	300	mm/s
	8) Speed Modes		0	3	5	decimal
	9) Repeat Limit Of Cancelled Print		0	50	250	mm
	External Print Signal Settings	10) External Print Signal Source	0	0	1	decimal
		11) External Print Signal Min. Duty Time	0	2	100	ms
	Internal Print Signal Settings	12) Internal Print Signal Activation	0	0	1	decimal
		13) Internal Print Signal Package Length	25	100	1000	mm
	Multi-Print Signal Settings	14) Multi-Print Signal Activation	0	0	1	decimal
		15) Multi-Print Signal Period	2	2	100	decimal
		16) Multi-Print Signal Package Length	25	100	1000	mm
	Ribbon Selection	17) Ribbon Selection	0	1	1	decimal
		Ribbon Outer Diameter	34	76	99	mm
		Ribbon Inner Diameter	25	33	35	mm
		Ribbon Length	300	600	1000	m
2	18) Tph Resistance Value		1050	1200	1475	Ohm
	19) Print Modes		0	0	8	decimal
	Ribbon Space Settings	20) Ribbon Unrewind Value	0	10	500	steps*mm/10
		21) Ribbon Rewind Value	0	10	500	steps*mm/10
	22) Printer Type		0	0	11	decimal
	23) Printer Cassette Type		0	0	1	decimal
	24) Tph Stroke (Pressing) Time		15	20	50	ms
	Ribbon Hanging Roller Settings	25) Ribbon Feeding Rubber Roller Diameter 1	20	40	60	mm
		26) Ribbon Feeding Rubber Roller Diameter 2	0	0	99	mm/100
	27)Tph Voltage Calibration		50	128	175	decimal
3	Ribbon Sensor Settings	29) Ribbon Broken Sensor Activation	0	0	3	decimal
		30) Ribbon Broken Sensor Limit	1	4	100	pulse
	Ribbon Tension Sensor Settings	31) Ribbon Tension Sensor Activation	0	0	1	decimal
		32) Ribbon Tension Sensor Limit	1	4	100	pulse
	Pre-Heating Settings	33) Pre-Heating Activation	0	0	1	decimal
		34) Pre-Heating Set Value	24	24	40	celsius
	35) Fuse Warning Setting		0	0	7	decimal
	36) Outputs Setting		0	0	7	decimal
4	Print Faulties Setting	37) Print Signal Faulties Activation	0	0	2	decimal
		38) Print Signal Faulty Limit Length	100	500	1500	mm
	39) Update Fault Activation		0	0	2	decimal
	40) Encoder Fault Activation		0	0	2	decimal
	41) Missing Print Fault Activation		0	0	2	decimal
	42) Tph Temperature Calibration		1	4	100	decimal

Table 6. Parameters

PARAMETER SETTINGS PAGE-1**ENCODER SETTINGS:**

A. Encoder Wheel Diameter: Encoder wheel showed in figure 12. Wheel diameter cannot be more than 60 mm, less than 40 mm. Standard encoder wheel diameter is 48 mm. If encoder diameter is changed, the value must be changed. Encoder diameter is used for speed calculation by printer.

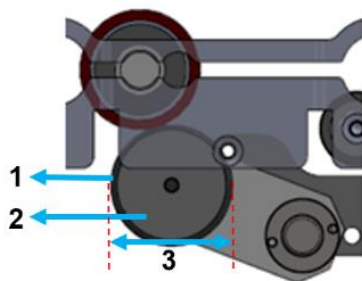


Figure 16. Encoder and Rubber Roller

No	Designation
1	ORING
2	ENCODER
3	ENCODER WHEEL DIAMETER

Table 7. Encoder and Rubber Roller Description

B. Encoder Pulse Number: Encoder pulse number is total pulse value per one cycle rotation of encoder. Encoder pulse number cannot be more than 2500 pulse and less than 500 pulse. Standard encoder pulse number is 1000 pulse. If encoder diameter is changed, the value must be changed. Printer may not operate properly if this value is changed without changing the encoder diameter.

C. Encoder Reading Period: The value is entered in which time intervals to read the value of the encoder pulse. The value can be reduced at high speeds, can be upgraded at low speeds. Standard encoder reading period is 2 ms. Encoder reading period cannot be more than 8 ms and less than 1 ms.

D. Encoder Rotation Direction: “0” or “1” is entered according to encoder rotation direction. “0” means that encoder is rotating anti-clockwise. “1” means that encoder is rotating clockwise. Default value is “1”.



Figure 17. Encoder Rotation Direction

No	Designation
1	CLOCKWISE ROTATION
2	ANTI-CLOCKWISE ROTATION

Table 8. Encoder Rotation Direction Descriptions

SLOW SPEED VIBRATION LIMIT:

Slow speed of vibration counter value is used to avoid printing cancelation at low speed. If vibration counter value exceed slow speed vibration limit or equal to the value, print process stops. With this setting, the printer continues printing if the packet speed slows down in short-time because of an unwanted situation occurs. Standard vibration limit is 20. The limit cannot be more than 200 and less than 0.

TPH HOLD (REST) WAITING VALUE:

Printer starts waiting after the TPH is in print position. When pulse number that comes from encoder exceed tph hold waiting value, printer prints. This setting helps to stabilize the head due to vibration. Waiting value cannot be more than 300 pulse and less than 0 pulse. Standard is 20 pulse for this parameter.

MINIMUM PRINT SPEED:

When package speed is less than this value while printer printing, printer cancels printing and stops. If the printer does not stop at instantaneous speed drops, it may be caused by the parameter “slow speed vibration limit”. Minimum print speed cannot be lower than 0 mm/s and upper than 300 mm/s. Standard minimum print speed is 25 mm/s.

SPEED MODES:

There are 6 speed modes;

Mode	Maximum Speed(mm/s)	Minimum Unprinted Space on Package(mm)
0	UP TO 300	25 mm
1	UP TO 400	35 mm
2	UP TO 500	45 mm
3	UP TO 600	55 mm
4	UP TO 800	65 mm
5	UP TO 1000	120 mm

Table 9. Speed Modes

The mode should be chosen according to the required criteria. “Minimum unprinted space on package” means that is distance between two templates. Default value is “3”.

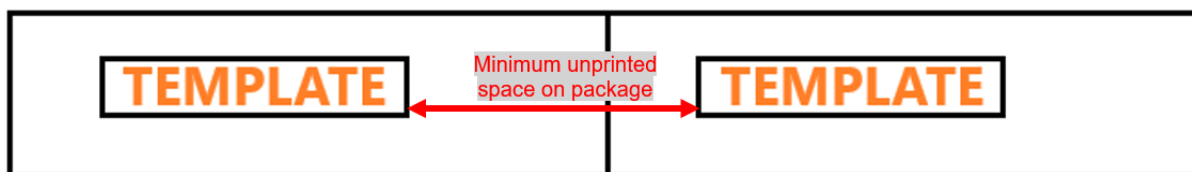


Figure 18. Maximum Unprinted Space on Package

REPEAT LIMIT OF CANCELLED PRINT:

After printing starts, repeat count starts. If printing cancel because of speed that lower than minimum speed in limit value, printer prints again cancelled printing after speed exceed minimum speed. The print must not exceed the repeat limit for reprinting. Otherwise the canceled print will not be printed again. Minimum limit is 0 mm, maximum limit is 250 mm and standard is 50 mm for this parameter. When repeat limit set 0 mm, printer repeat function is cancelled.

EXTERNAL PRINT SIGNAL SETTINGS:

A. External Print Signal Source: This parameter is used to choose contact type. If “0” is chosen, contact 1 is active and contact 2 is passive. If “1” is chosen, contact 1 is passive and contact 2 is active. Default value is “0”.

NOTE: External signal, internal signal and multi-print signal cannot be used at the same time. Only one of them can be used.

If contact 1 is selected, +24 V DC must be supplied to the contact 1 terminal on the contact wire. The GND of 24 V DC must also be connected to contact 1 GND on the contact wire. The same procedure is similar for contact 2. If contact 2 is selected, +24 V DC is connected to contact 2, and GND of 24 V is connected to contact 2 GND.

B. External Print Signal Min. Duty Time: Contact signal duty time is not smaller than this value to print. Otherwise, no printing process. Standard duty time is 2 ms. The duty time cannot be more than 100 ms and less than 0 ms. This parameter is used to prevent undesired signal. Following sample is valid for the value is equal to 2 ms.

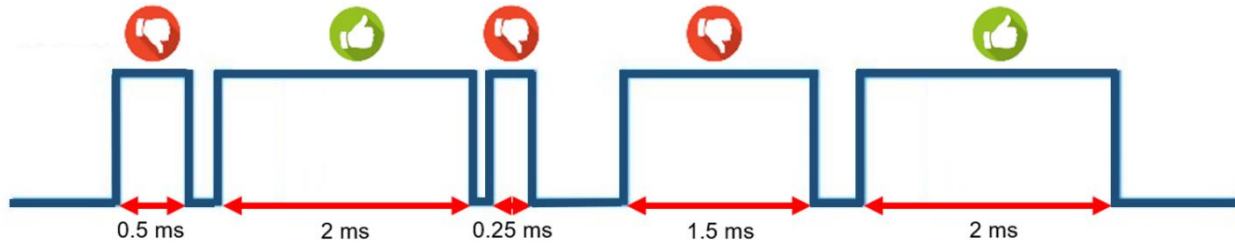


Figure 19. External Print Signal Minimum Duty Time Sample

INTERNAL PRINT SIGNAL SETTINGS:

A. Internal Print Signal Activation: This parameter is used to activate internal print function. If “0” is chosen, the function is passive. If “1” is chosen, the function is active. Default value is “0”.

NOTE: External signal, internal signal and multi-print signal cannot be used at the same time. Only one of them can be used.

B. Internal Print Signal Package Length: Printer takes contact signal from internal signal source depend on pack length if the parameter is activated. Internal print signal package length is distance of two print out. The length have to be bigger than template length + minimum unprinted space on package (See “speed modes parameter” for more detail about minimum unprinted space on package). Internal print signal minimum package length cannot be lower than 25 mm and upper than 1000 mm. Standard length is 100 mm. Following sample is valid for the value is equal to 200.

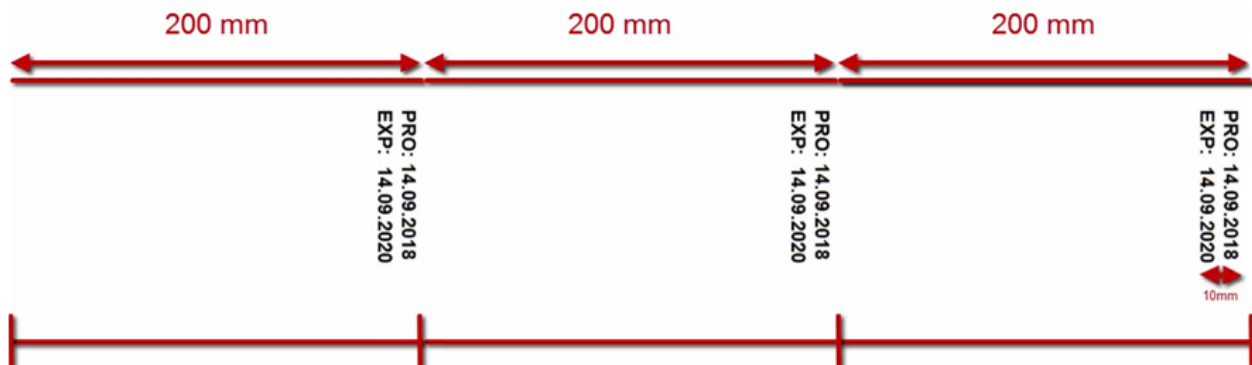


Figure 20. Internal Print Sample

MULTI-PRINT SIGNAL SETTINGS:

A. Multi-Print Signal Activation: This parameter is used to activate multi print function. If “0” is chosen, the function is passive. If “1” is chosen, the function is active. Default value is “0”.

NOTE: External signal, internal signal and multi-print signal cannot be used at the same time. Only one of them can be used.

B. Multi-Print Signal Period: Multi-Print signal period is number of print in an external print signal. The period cannot be more 100 and less than 2. Standard is 2 for this parameter.

C. Multi-Print Signal Package Length: Printer takes contact signal from trigger signal source depend on package length if the parameter is activated. Multi-Print signal package length is distance of two contact signal. The package length parameter cannot be more than 1000 mm and less than 25 mm. Standard is 100 mm for this parameter.

The following sample is valid for multi-print signal period is equal to 3 and multi-print signal package length is 200 mm.

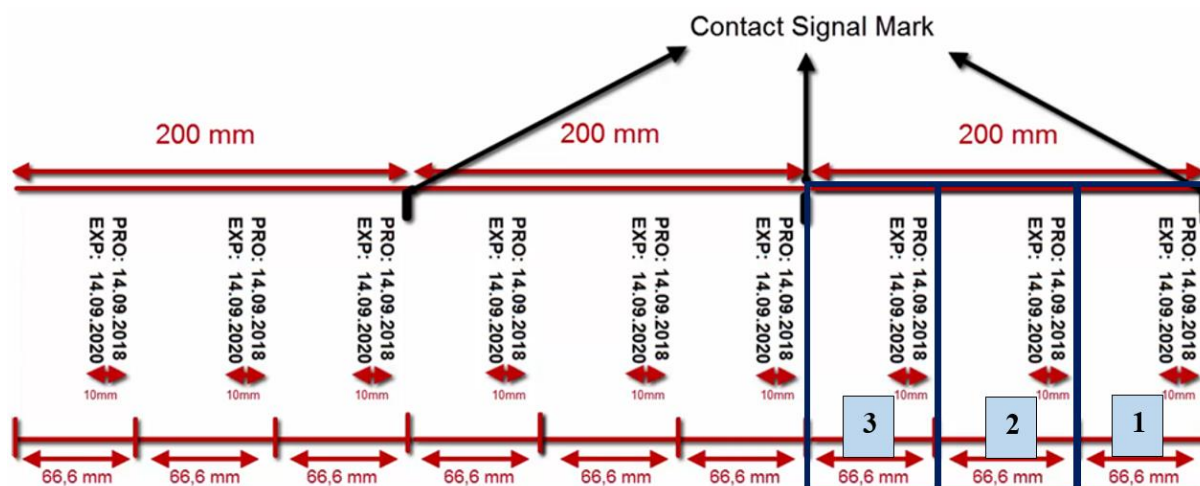


Figure 21. Multi-Print Sample

RIBBON SELECTION:

A. Ribbon Selection: There are two modes. Mode 0 is for high speed ribbon and mode 1 is for normal speed ribbon. The mode can be changed according to the type of ribbon used. Mode 0 is selected if the ribbon used supports high speed. Default mode is 1.

B. Ribbon Outer Diameter: Ribbon outer diameter cannot be more than 99 mm and less than 34 mm. Standard diameter is 76 mm.

C. Ribbon Inner Diameter: Ribbon inner diameter cannot be more than 35 mm and less than 25 mm. Standard diameter is 33 mm. When measuring the inner diameter of the ribbon, neglect the roller it is wound on.

D. Ribbon Length: Ribbon length cannot be more than 1000 m and less than 300 m. Standard diameter is 600 m. Ribbon length cannot be selected more than 600 m if printer is not XL model.

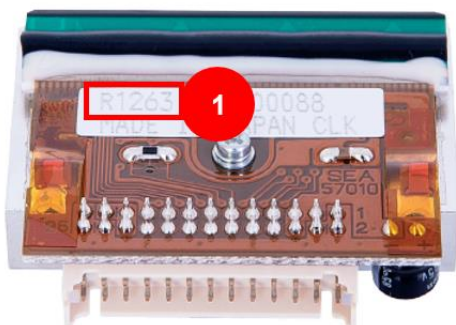
PARAMETER SETTINGS PAGE-2**TPH RESISTANCE VALUE:**

Figure 22. Thermal Print Head

Thermal print head resistance value(R) written on TPH label. The resistance cannot be lower than 1050 Ω and upper than 1475 Ω .

No	Designation
1	TPH RESISTANCE VALUE

Table 10. Thermal Print Head Description

PRINT MODES:

There are 9 print modes. These modes change according to dot for a pixel, maximum speed and ribbon type.

Mode	Pixel	Ribbon Type	Maximum Speed
0	ONE DOT FOR A PIXEL	WAX-RESIN RIBBON	1000
1	ONE DOT FOR A PIXEL	RESIN RIBBON	550
2	Not used	Not used	Not used
3	DOUBLE DOT FOR A PIXEL	WAX-RESIN RIBBON	1000
4	DOUBLE DOT FOR A PIXEL	RESIN RIBBON	550
5	Not used	Not used	Not used
6	TRIPLE DOT FOR A PIXEL	WAX-RESIN RIBBON	1000
7	TRIPLE DOT FOR A PIXEL	RESIN RIBBON	550
8	Not used	Not used	Not used

Table 11. Print Modes

The mode should be chosen according to the required criteria. Since the printer automatically adjusts to the selected mode, the mode must be selected according to the ribbon type used. Otherwise, print quality may be poor. Mode 0-3-6 are used for wax-resin ribbon and have same print quality. Mode 1-4-7 are used for resin ribbon and have same print quality. Standard mode is "0".

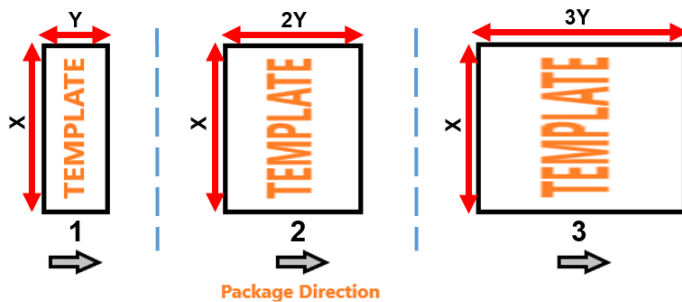


Figure 23. Dot for a Pixel Samples

No	Designation (Template Dimensions)
1	ONE DOT FOR A PIXEL
2	DOUBLE DOT FOR A PIXEL
3	TRIPLE DOT FOR A PIXEL

Table 12. Dot for a Pixel Types

RIBBON SPACE SETTING:

A. Ribbon Unrewind Value: This value decreases the gap of two print-out on ribbon. Each 10 motor pulses affect 1mm. The wind value cannot be less than 0 step*mm/100 and more than 500 step*mm/100. Standard is 10 step*mm/100 for this parameter.

B. Ribbon Rewind Value: This value increases the gap of two print-out on ribbon. Each 10 motor pulses affect 1mm. The rewind value must be more than 0 step*mm/100 and less than 500 step*mm/100. Standard is 10 step*mm/100 for this parameter.

PRINTER TYPE:

Mode	Printer Type	Mode	Printer Type	Mode	Printer Type
0	32 mm	4	32 mm XL	8	32 mm AB
1	53 mm	5	53 mm XL	9	53 mm AB
2	107 mm	6	107 mm XL	10	107 mm AB
3	128 mm	7	128 mm XL	11	128 mm AB

Table 13. TPH Type Modes

This parameter is used to declare to system the type of printer used. There are 12 printer type and default value is entered according to buying printer type.

PRINTER CASSETTE TYPE:

This parameter is used to declare to system whether the device is cassette. If “0” is entered, that’s mean the device has cassette. If “1” is entered, that’s mean the device has no cassette. Standard type is entered according to buying machine cassette type.

TPH STROKE (PRESSING) TIME:

This parameter is used to adjust TPH reach time from piston position to rubber roller position after contact signal was received. The time cannot be more than 50 ms and less than 15 ms. Standard is 20 ms for this parameter

RIBBON HANGING ROLLER SETTING:

A. Ribbon Rubber Roller Diameter 1: Diameter of rubber roller should be entered. The diameter cannot be more than 60 mm and less than 20 mm. Standard is 40 mm for this parameter.

B. Ribbon Rubber Roller Diameter 2: Percentage value of rubber roller diameter should be entered. The diameter cannot be more than 99 mm/100 and less than 0 mm/100. Standard is 0 mm/100 for this parameter.

For example; if ribbon feeding rubber roller diameter is 30.50 mm, ribbon feeding rubber roller diameter 1 should be entered 30 and ribbon feeding rubber roller diameter 2 should be entered 50.

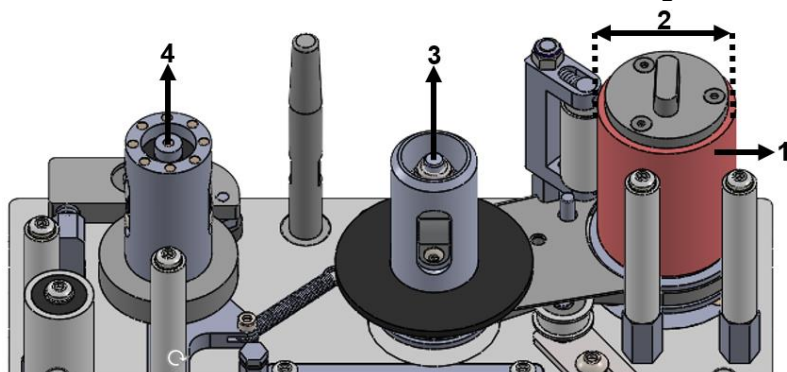


Figure 24. Rubber Roller Diameter Representation

No	Designation
1	RIBBON RUBBER ROLLER
2	RIBBON RUBBER ROLLER DIAMETER
3	USED RIBBON
4	UNUSED RIBBON

Table 14. Description of Rubber Roller Representation

TPH VOLTAGE CALIBRATION:

Mainboard factory setting should not change without technical personal. TPH resistance value must be “1200” and ribbon thickness setting must be “1” for calibration. TPH voltage can adjust as 24V with this value. TP3 and TPGND test points on the mainboard are used to measure voltage. The measurement is made by placing the “+” end of the multimeter to the TP3 and the “-” end to the TPGND. The calibration value cannot be less than 50 and more than 175. Standard is 128 for this parameter.

PARAMETER SETTINGS PAGE-3**RIBBON SENSOR SETTING:**

A. Ribbon Broken Sensor Activation: There are 4 modes and default value is “0”;

Mode		
0	RIBBON BREAK ACTIVE	RIBBON NOT FOUND ACTIVE WITH MAGNETS
1	RIBBON BREAK PASSIVE	RIBBON NOT FOUND ACTIVE WITH MAGNETS
2	RIBBON BREAK PASSIVE	RIBBON NOT FOUND PASSIVE WITH MAGNETS
3	RIBBON BREAK ACTIVE	RIBBON NOT FOUND ACTIVE WITH SWITCH

Table 15. Ribbon Broken Sensor Activation Modes

B. Ribbon Broken Sensor Limit: When mode 0 is selected, ribbon broken alarm occur after sensor limit value. The limit cannot be more than 100 pulse and less than 1 pulse. Standard limit is 4 pulse for this parameter.

RIBBON TANSION SENSOR SETTING:

A. Ribbon Tension Sensor Activation: This parameter is used to activate ribbon tension sensor. If “0” is chosen, the sensor is active with magnets. If “1” is chosen, the sensor is passive. Default value is “0”.

B. Ribbon Tension Sensor Limit: When sensor is activated, ribbon tension alarm occur after limit of magnets value. The limit cannot be more than 100 pulse and less than 1 pulse. Standard limit is 4 pulse for this parameter.

PRE-HEATING SETTING:

A. Pre-Heating Activation: This parameter is used to activate pre-heating. If “0” is chosen, the pre-heating is active. If “1” is chosen, pre-heating is passive. Default value is “0”.

B. Pre-Heating Set Value: Pre-heating system controls temperature. This value determine minimum TPH temperature. The value cannot be more than 40 °C and less than 24 °C. Standard value is 24 °C.

FUSE WARNING SETTING:

Default value is “0” and there are 8 modes (MB: Main Board - MD: Motor Driver).

Mode			
0	MB FUSE WARNING ACTIVE	MD FUSE WARNING ACTIVE	TPH ERROR ACTIVE
1	MB FUSE WARNING PASSIVE	MD FUSE WARNING ACTIVE	TPH ERROR ACTIVE
2	MB FUSE WARNING ACTIVE	MD FUSE WARNING PASSIVE	TPH ERROR ACTIVE
3	MB FUSE WARNING PASSIVE	MD FUSE WARNING PASSIVE	TPH ERROR ACTIVE
4	MB FUSE WARNING ACTIVE	MD FUSE WARNING ACTIVE	TPH ERROR PASSIVE
5	MB FUSE WARNING PASSIVE	MD FUSE WARNING ACTIVE	TPH ERROR PASSIVE
6	MB FUSE WARNING ACTIVE	MD FUSE WARNING PASSIVE	TPH ERROR PASSIVE
7	MB FUSE WARNING PASSIVE	MD FUSE WARNING PASSIVE	TPH ERROR PASSIVE

Table 16. Fuse Modes

If MB fuse warning is activated, printer gives error when mainboard fuse is damaged. In this case the fuse on mainboard should be controlled and changed if necessary. In changing situation, pay attention to the value of the fuse. Detailed information on the location and value of the mainboard fuse is discussed in section 4.1.1.

If MD fuse warning is activated, printer gives error when motor driver fuse is damaged. In this case the fuse on motor driver should be controlled and changed if necessary. In changing situation, pay attention to the value of the fuse. Detailed information on the location and value of the motor driver fuse is discussed in section 4.1.2.

If TPH error is activated, there are 3 situation that cause to error.

1. When TPH temperature is less than 1 °C and more than 55 °C.
2. When TPH is not installed.
3. When TPH cable is broken or not fully inserted.

OUTPUTS SETTING:

Default value is “0” and there are 8 fault output modes;

Mode		
0	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL PASSIVE(FC)
1	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL PASSIVE(FC)
2	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL ACTIVE(FC)
3	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL ACTIVE(FC)
4	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL PASSIVE(HC)
5	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL PASSIVE(HC)
6	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL ACTIVE(HC)
7	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL ACTIVE(HC)

Table 17. Alarm Modes

- FC means full cycle. Printer gives busy signal while TPH is in print process and during come back to initial state.
- HC means half cycle. Printer gives busy signal while TPH is in print process but doesn't give busy signal during come back to initial state.
- ERROR times mean an error has occurred in the system like ribbon break etc.
- Passive time means the printer is at stop position not print position.

PARAMETER SETTINGS PAGE-4**PRINT FAULTIES SETTINGS:**

A. Print Signal Faulties Activation: This parameter is used to adjust print signal error is active or passive. When adjusting the print signal error setting, warning and fault out settings are also made. Warning means if there is an error, the error appears on screen. Fault out means determines whether the system gives error output even if there is fault warning. Default value is “0” and there are 3 modes;

Mode	Warning	Fault Out	Print Signal Error
0	PASSIVE	PASSIVE	PASSIVE
1	ACTIVE	ACTIVE	ACTIVE
2	ACTIVE	PASSIVE	ACTIVE

Table 18. Print Signal Fault Modes

Warning: If warning property is activated, warning related activated print signal error occur on screen. When warning is not activated, even if error occurs user does not see error warning on screen.

Fault out: If fault out property is activated, printer gives error output related activated print signal error. When fault out is not activated, even if error occurs printer does not give output.

Print Signal Error: If print signal error property is activated, printer gives printer signal error when contact signal does not received during the time entered in print signal faulty limit length parameter.

B. Print Signal Faulty Limit Length: When print signal error is activated, printer wait print signal to not give error as print signal fault limit. If print signal is not received in print signal faulty limit length, printer gives print signal error. The length cannot be more than 1500 mm and less than 100 mm. Standard length is 500 for this parameter.

UPDATE FAULT ACTIVATION:

This parameter is used to adjust update error is active or passive. When adjusting the update error setting, warning and fault out settings are also made. Warning means if there is an error, the error appears on screen. Fault out means determines whether the system gives error output even if there is fault warning. Default value is "0" and there are 3 modes;

Mode	Warning	Fault Out	Update Error
0	PASSIVE	PASSIVE	PASSIVE
1	ACTIVE	ACTIVE	ACTIVE
2	ACTIVE	PASSIVE	ACTIVE

Table 19. Update Fault Modes

Warning: If warning property is activated, warning related activated update error occur on screen. When warning is not activated, even if error occurs user does not see error warning on screen.

Fault out: If fault out property is activated, printer gives error output related activated update error. When fault out is not activated, even if error occurs printer does not give output.

Update Error: If update error property is activated, printer gives update error when variable template doesn't change. Variable template is like number of package, serial number that specific to each product etc.

ENCODER FAULT ACTIVATION:

This parameter is used to adjust encoder error is active or passive. When adjusting the encoder error setting, warning and fault out settings are also made. Warning means if there is an error, the error appears on screen. Fault out means determines whether the system gives error output even if there is fault warning. Default value is "0" and there are 3 modes;

Mode	Warning	Fault Out	Encoder Error
0	PASSIVE	PASSIVE	PASSIVE
1	ACTIVE	ACTIVE	ACTIVE
2	ACTIVE	PASSIVE	ACTIVE

Table 20. Encoder Fault Modes

Warning: If warning property is activated, warning related activated encoder error occur on screen. When warning is not activated, even if error occurs user does not see error warning on screen.

Fault out: If fault out property is activated, printer gives error output related activated encoder error. When fault out is not activated, even if error occurs printer does not give output.

Encoder Error: If encoder error property is activated, printer gives encoder error when encoder doesn't rotate or rotate wrong direction.

MISSING PRINT FAULT ACTIVATION:

This parameter is used to adjust missing print error is active or passive. When adjusting the missing print error setting, warning and fault out settings are also made. Warning means if there is an error, the error appears on screen. Fault out means determines whether the system gives error output even if there is fault warning. Default value is "0" and there are 3 modes;

Mode	Warning	Fault Out	Missing Print Error
0	PASSIVE	PASSIVE	PASSIVE
1	ACTIVE	ACTIVE	ACTIVE
2	ACTIVE	PASSIVE	ACTIVE

Table 21. Missing Print Fault Modes

Warning: If warning property is activated, warning related activated missing print error occur on screen. When warning is not activated, even if error occurs user does not see error warning on screen.

Fault out: If fault out property is activated, printer gives error output related activated missing print error. When fault out is not activated, even if error occurs printer does not give output.

Missing Print Error: If missing print error property is activated, printer gives missing print error when the template is not printed completely.

TPH TEMPERATURE CALIBRATION:

This parameter is used to read print head temperature calibration value. The calibration value cannot be less than 1 and more than 100. Standard is 4 for this parameter.

1.2 ERRORS

1.2.1 COMMUNICATION ERRORS



After cable installation; turn power on/off switch to “ON” mode. When you turn the power switch, firstly open screen will come and if all the connections well, screen goes to next window. If there is a communication problem, screen shows message on the left. This error is not dependent on to the parameter, so it cannot be set to be passive or active with parameter.

When the error occurs, check steps in below;

- The communication cable between the screen and the printer must be fully inserted into the sockets and the cable must be unbroken.
- The communication cable between mainboard and screen board must be fully inserted into the sockets and the cable must be unbroken.
- The screen board must be unbroken (there should be no burn or discharge marks around any component).
- The little board in the screen must be fully inserted into the socket.

If the communication between the display and the printer suddenly stops after the printer has started, the following warning appears

Rs422 communication error

When the error occurs, check steps in below;

- The communication cable between the screen and the printer must be fully inserted into the sockets and the cable must be unbroken.
- The communication cable between mainboard and screen board must be fully inserted into the sockets and the cable must be unbroken.

1.2.2 RIBBON BREAK/RIBBON NOT FOUND ERROR

The machine has a switch that detects the ribbon break and sensors that detect the absence of ribbon in the system. These elements inform the printer in the event of any problem. If there is a ribbon break problem, screen shows message on the bottom.

Existing Ribbon broken. Please change the Ribbon.

When ribbon break error occurs, check steps in below;

- Air pressure should not exceed 2–2.5 bars (maximum).
- Ribbon mounting must be correct. If not, there can be a ribbon tension problem. It must be checked according to the instructions on the cartridge cover.
- Check the ribbon tension. If it is not good, then determine the reason and correct the problem.
- The ribbon may be wrapped around the rubber roll. Investigate to see if there is any problem with ribbon conveyance.
- If the thermal print head printing edge is not parallel to the printing surface, then the ribbon may break.
- The ribbon broken sensor cable on the mainboard must be fully seated and not broken.

If there is a ribbon not found problem, screen shows message on the bottom.

Ribbon not found. Please insert ribbon.

When ribbon not found error occurs, check steps in below;

- Ribbon should be replaced according to the instructions on the cartridge cover.
- The ribbon broken sensor cable and switch cable on the mainboard must be fully seated and not broken.

Activation of ribbon break and ribbon not found errors can be adjust in the parameter menu. Ribbon sensor setting/Ribbon broken sensor activation parameter dedicated to this work.

1.2.3 TPH ERROR

Thermal print head (TPH) is the part of the printer used to print. Tph-related problems may occur in some cases. In this case the warning message on the bottom will appear.

Thermal print head error. Please check below steps.

- 1- Print head cable must be inserted well.
 - 2- Print head temperature must be between +1 and +55 °C.
 - 3- Print head must be robust.
 - 4- Print head cable must be robust.
- If problem continue, please contact with manufacturer.**

Activation of TPH error can be adjust in the parameter menu. Fuse warning setting parameter dedicated to this work.

When TPH error occurs, check steps in below;

- Tph must be installed in the system.
- Tph cable should not be broken
- Tph cable must be fully inserted into the TPH cable connection board socket.
- The cable must be fully inserted into the print head cable socket.

- Optimal temperature must be 24 °C. Pre-heating setting/pre-heating activation parameter should be “0” for automatically heating.

1.2.4 MAINBOARD/MOTOR DRIVER FUSE ERROR

Fuses are used in electronic circuit boards to protect devices after a certain current. If there is a mainboard fuse problem, screen shows message on the bottom.

Main board fuse is broken. Please contact manufacturer.

If there is a motor driver problem, screen shows message on the bottom.

Motor driver fuse is broken. Please contact manufacturer.

When fuse error occurs, check steps in below;

- Make sure the fuses are not damaged. The location of the mainboard fuse is shown in section 4.1.1 and the location of the motor driver board fuse is shown in section 4.1.2. If there is damage, it must be replaced.

Activation of fuse errors can be adjust in the parameter menu. Fuse warning setting parameter dedicated to this work. The fuses can be checked for damage in the control menu (for detail see section 1.1.4.2).

1.2.5 MISSING PRINT ERROR

Printer gives missing error when printing process is not complete. If there is a missing problem, screen shows message on the bottom.

Missing print error

When missing error occurs, check steps in below;

- Make sure the encoder touches the cylinder.
- Prevent any vibration caused by the packaging machine.
- Make sure the pistons are working correctly.
- Make sure the head is in the correct position (more detail see section 3.5.1).
- Air pressure should not exceed 2–2.5 bars (maximum).

Activation of missing print error can be adjust in the parameter menu. Missing print fault activation parameter dedicated to this work.

1.2.6 UPDATE ERROR

Some templates change with each print. These are called variable templates. Variable template is like number of package, serial number that specific to each product etc. Printer gives update error when the printer prints the same print twice. If there is an update problem, screen shows message on the bottom.

Update error

When update error occurs, check steps in below;

- Be sure to use a variable template. If the template does not have this property, update error property should be turned off.
- Communication may be slow if there are too many variables in the same template. In this case the print speed should be reduced.

Activation of update error can be adjust in the parameter menu. Update fault activation parameter dedicated to this work.

1.2.7 PRINT SIGNAL ERROR

The printer needs a contact signal to print. Printer gives print signal error if the printer does not receive contact signal within the value entered in print faults setting/print signal faulty limit length parameter after switching the printer to print mode. If there is a print signal problem, screen shows message on the bottom.

Print signal error

When print signal error occurs, check steps in below;

- The value entered in the print faults setting/print signal faulty limit length parameter must be greater than the packet length used.
- The contact cable must be fully inserted into the sockets and the cable must be unbroken.
- Check the contact signal to the printer.
- The contact board must be unbroken (there should be no burn or discharge marks around any component).

Activation of print signal error can be adjust in the parameter menu. Print faults setting/Print faults activations parameter dedicated to this work.

1.2.8 ENCODER ERROR

Encoder is used as hardware to inform the printer of the speed of the packaging machine. Printer gives encoder error when the encoder doesn't rotate or rotate wrong direction. If there is an encoder problem, screen shows message on the bottom.

Encoder doesn't rotating or does rotating inverse. Please check encoder connection.

When encoder error occurs, check steps in below;

- Make sure encoder cable fully inserted.
- Make sure the encoder touches the roller under the thermal head.
- Be sure that the direction of rotation of the encoder is entered correctly in encoder settings/encoder rotation direction parameter.

Activation of encoder error can be adjust in the parameter menu. Encoder fault activation parameter dedicated to this work.

1.2.9 TEMPLATE ERROR

When the machine is first turned on, the printer does not have a template under normal state. The screen shows following warning to install a template on the printer.

Please install the template.

When template error occurs, check steps in below;

- Make sure template is installed.

1.2.10 MEMORY ERROR

If there is a problem ram on the mainboard, screen shows warning on the bottom.

Memory error

When memory error occurs, check steps in below;

- Ram may be corrupted. Please contact manufacturer.

1.2.11 MOTOR DRIVER COMMUNICATION ERROR

When there is a communication problem between motor driver and mainboard, screen shows following warning.

Motor driver communication error

When motor driver communication error occurs, check steps in below;

- Make sure communication cable between motor driver and mainboard is fully inserted.

1.2.12 VOLTAGE ADJUSTMENT ERROR

When TPH voltage is lower than 24 °C, screen shows warning on the bottom.

Voltage adjustment error

When voltage adjustment error occurs, check steps in below;

- TPH resistance value is entered "1200" on parameter menu.
- Ribbon thickness setting is entered "1".
- TP3 and TPGND test points on the mainboard are measured. The measurement is made by placing the "+" end of the multimeter to the TP3 and the "-" end to the TPGND.
- TPH voltage calibration value is changed until the value on the meter indicates 24 volts.
- TPH resistance value is rechanged to original TPH resistance.

1.2.13 RIBBON TENSION ERROR

The ribbon must has a certain tension for the printer to print. When the ribbon has not optimal tension, screen shows warning on the bottom.

Ribbon tension error

When ribbon tension error occurs, check steps in below;

- Air pressure should not exceed 2–2.5 bars (maximum).
- Make sure the encoder is working correctly.
- Prevent any vibration caused by the packaging machine.
- Make sure the pistons are working correctly.
- Make sure the head is in the correct position (more detail see section 3.5.1).
- Make sure TPH temperature is 24 °C.

1.2.14 COVER ERROR

When the printer cover is not fully inserted, the following message is shown on the screen.

Printer Cover is Open. Please Close the Cover.

When cover error occurs, check steps in below;

- Gently pull the cover outward and push it straight into the printer.

1.3 CASSETTE LED DESCRIPTIONS

There are LEDs on the cassette which provide information about various situations. The color of the LEDs may vary from product to product. The reference point is the location of the printer. On a flat-installed printer, the leftmost LED indicates communication, the middle LED indicates fault, the rightmost LED indicates contact.

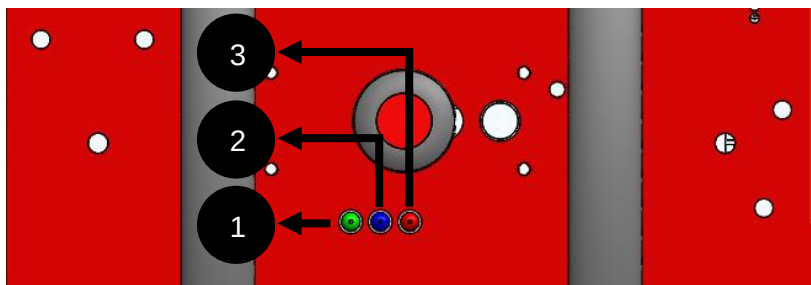


Figure 25. Information LEDs on Cassette

No	Designation
1	COMMUNICATION LED
2	FAULT LED
3	CONTACT LED

Table 22. LEDs Descriptions on Cassette

1. Communication LED: This LED indicates communication between the screen and the printer when the machine is first turned on. If the LED is blinking during opening, there is no problem in communication. There is a led used for the same operation on the mainboard. In order to control that led, it is necessary to remove the case of the machine. The LED on the cassette is designed to eliminate this difficulty.

This LED is used for another operation after the machine has completed the opening process. Another operation is that the LED is on as long as the thermal head is in the printing position.

2. Fault LED: This LED is on when the printer gives an error. For proper operation of the LED, it should be ensured that the desired errors are activated in the parameters section.

3. Contact LED: This LED lights up when the printer receives a contact signal. The LED is designed to check whether the contact signal comes correctly. This led only works when contact 1 input is used. If the contact is given from the contact 2 input, the led will not operate.

2

ELECTRICAL DIAGRAMS

2.1 MACHINE ELECTRONIC COMPONENTS AND BOARDS CONNECTION

Basically, there are 4 boards in the printer;

- Mainboard
- Motor driver board
- Screen communication board (power board)
- Contact board

Besides, there is a connection board for TPH cable. Also there are 2 boards in screen;

- Screen board
- COM board

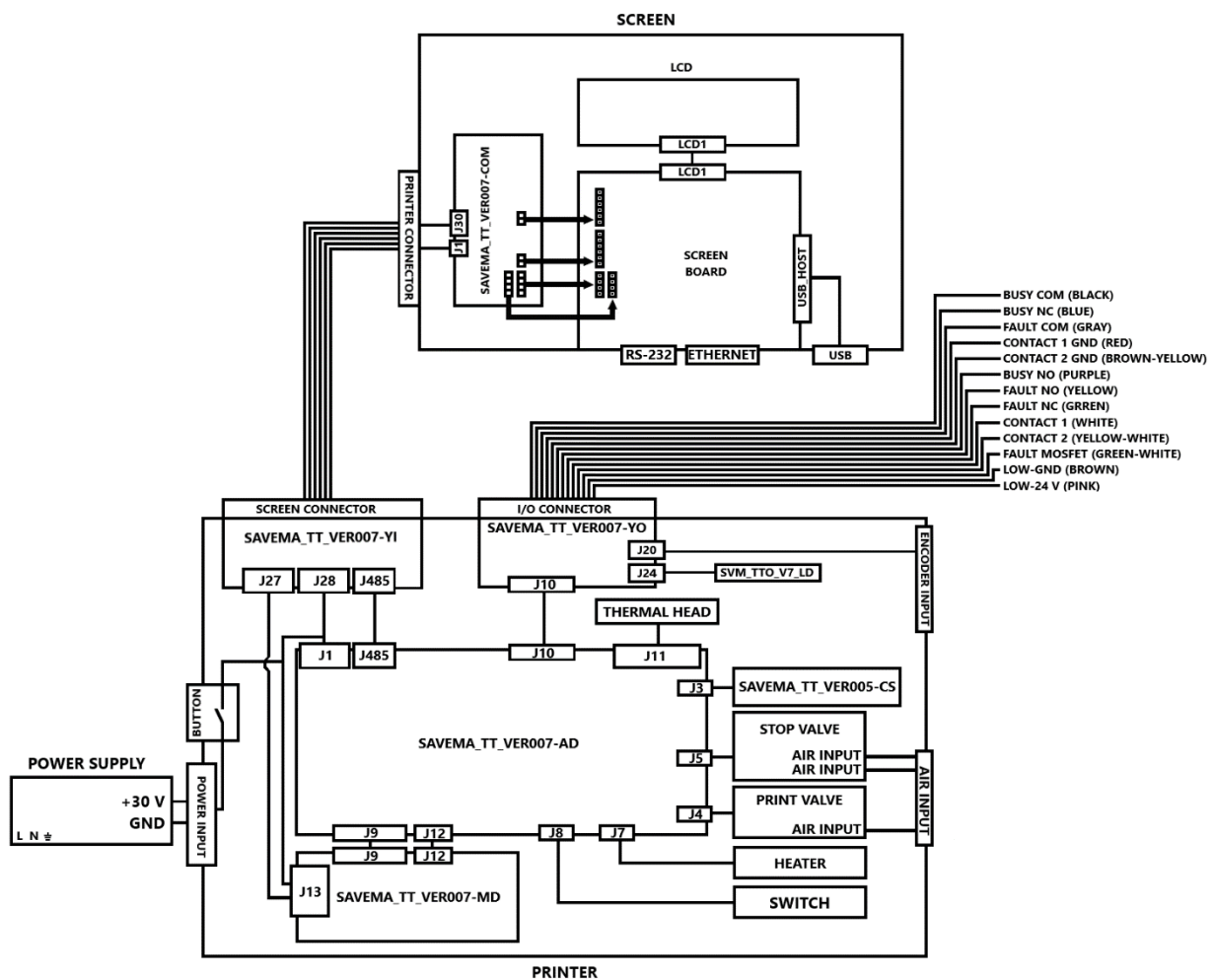


Figure 26. General Electrical Connections

2.2 PACK MACHINE CONNECTIONS

Contact signal is required for the printer to print. If this contact signal is to be supplied from an external source (such as a photocell or relay), attention must be paid to the type of source, ie whether it is PNP or NPN. Once the type of source has been determined, the connection can be made as shown below.

In addition, the printer generates fault and busy signals to be used in the package machine. The connections of these signals to the packaging machine are shown below.

There are 2 contact inputs for the printer. These two inputs cannot be used at the same time. Contact 2 is designed to be an alternative when the contact 1 input fails. Which contact to use must be activated in the parameters section. External print signal settings / external print signal source parameter dedicated to this work. Also, the printer cannot detect the signal from an inactive contact. In addition, if contact 1 is used, contact 1 GND must be used as GND. If contact 2 is used, contact 2 GND must be used as GND.

2.2.1 PNP CONNECTIONS

Photocell sensors have two different types, PNP or NPN. The connection diagram varies depending on the type of sensor. This is a point to be considered for the smooth operation of the system. The PNP type sensors trigger the positive voltage. Connection must be made according to this rule.

The relays have three ports; COM, NO, NC. When the relay is triggered by means of terminals A1+ and A2-, the signal given from terminal COM can be used from terminal NO. When the A1+ and A2- terminals are not triggered, the signal from the COM terminal is transmitted to the NC terminal. For example, if 24 V is connected to the COM terminal and the relay is triggered, there will be 24 V at the NO end and no signal at the NC. Similarly, if 24 V is connected to the COM terminal and the relay is not triggered, there will be 24 V at the NC terminal, and no signal will be at the terminal NO.

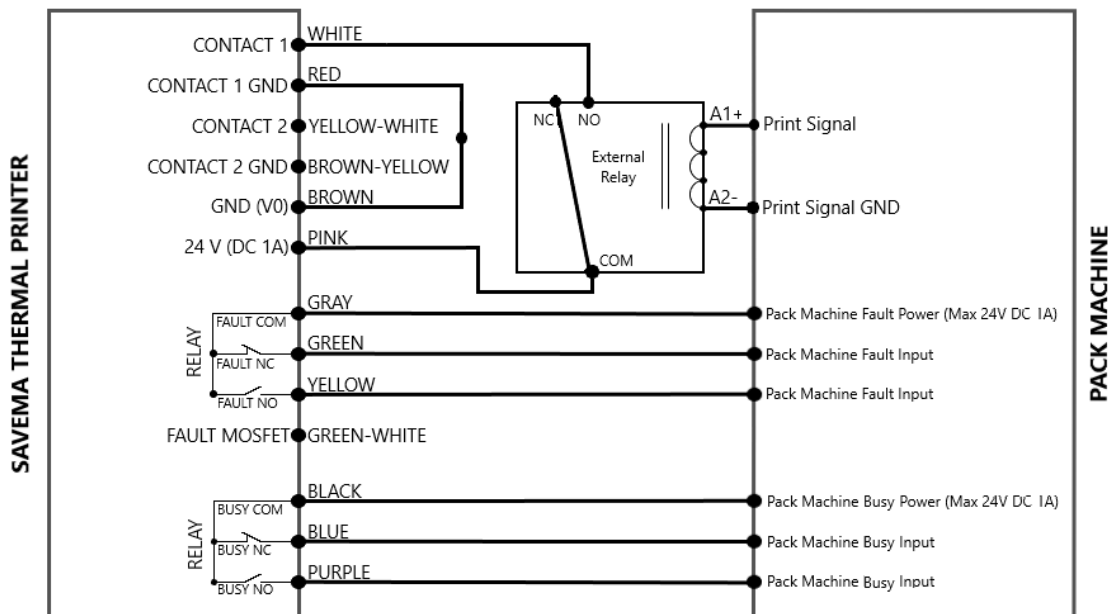


Figure 27. PNP Relay Connections

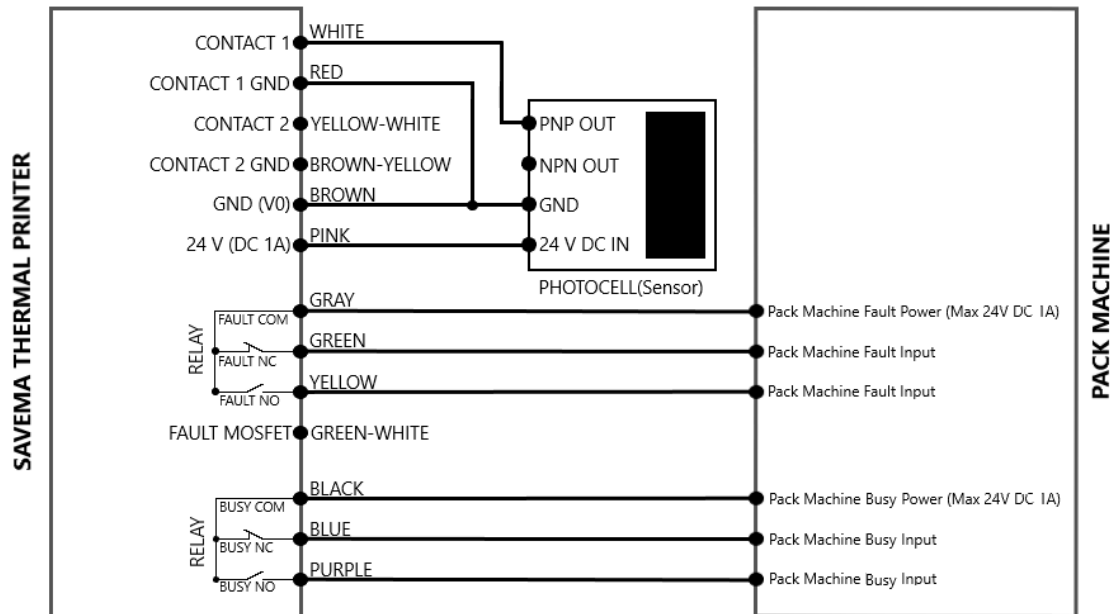


Figure 28. PNP PHOTOCELL Connections

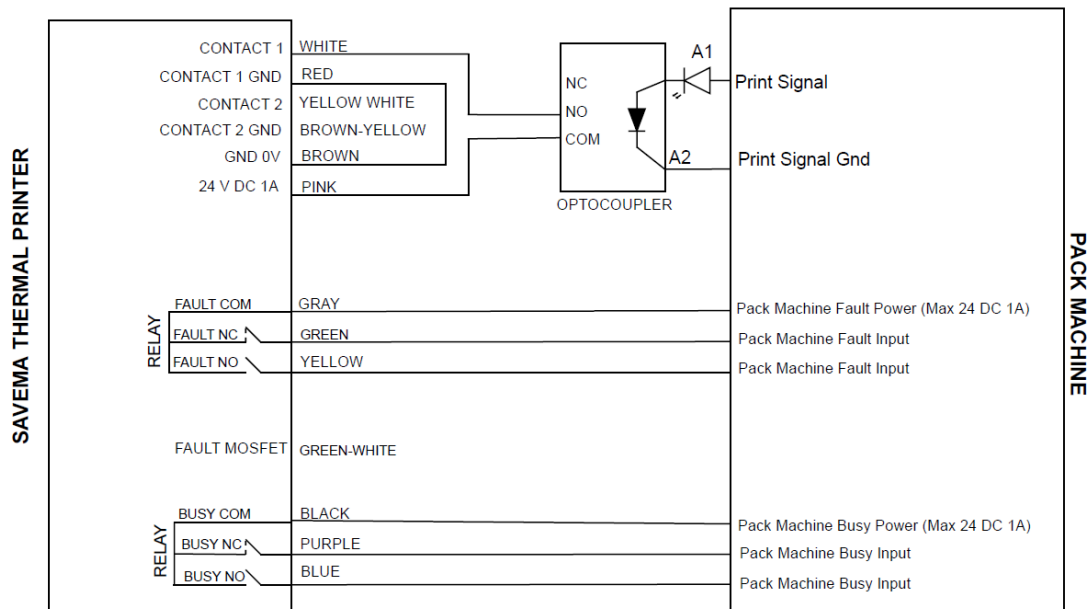


Figure 29 PNP Optocoupler Connection

2.2.2 NPN CONNECTIONS

Photocell sensors have two different types, PNP or NPN. The connection diagram varies depending on the type of sensor. This is a point to be considered for the smooth operation of the system. The NPN type sensors trigger the negative/neutral voltage. Connection must be made according to this rule.

The relays have three ports; COM, NO, NC. When the relay is triggered by means of terminals A1+ and A2-, the signal given from terminal COM can be used from terminal NO. When the A1+ and A2- terminals are not triggered, the signal from the COM terminal is transmitted to the NC terminal. For example, if GND is connected to the COM terminal and the relay is triggered, there will be 0 V at the NO end and no signal at the NC. Similarly, if GND is connected to the COM terminal and the relay is not triggered, there will be 0 V at the NC terminal, and no signal will be at the terminal NO.

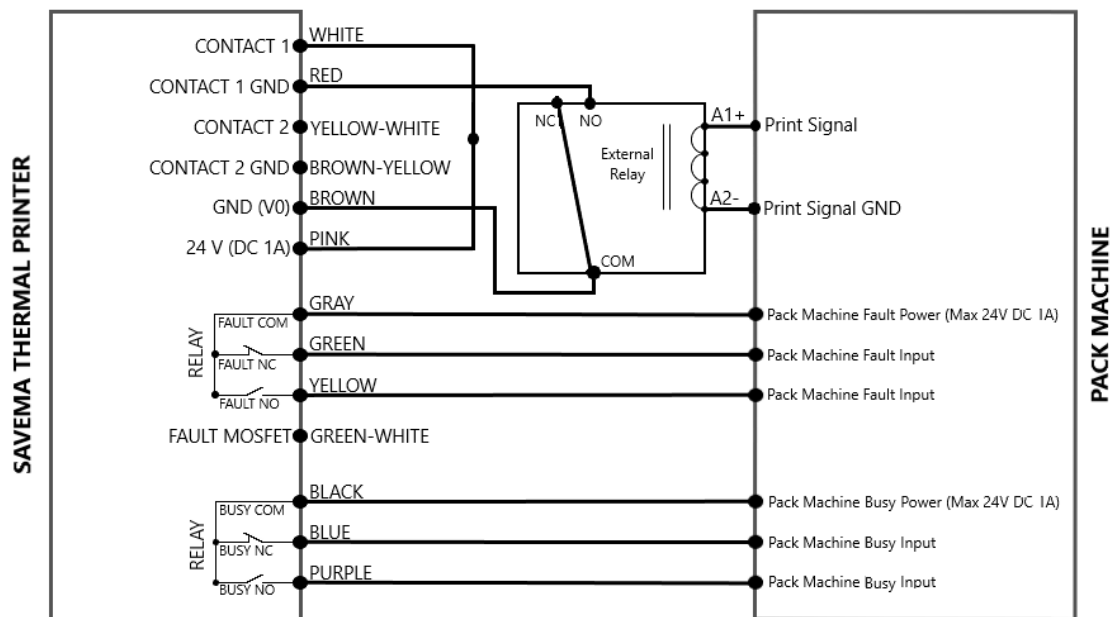


Figure 30. NPN Relay Connections

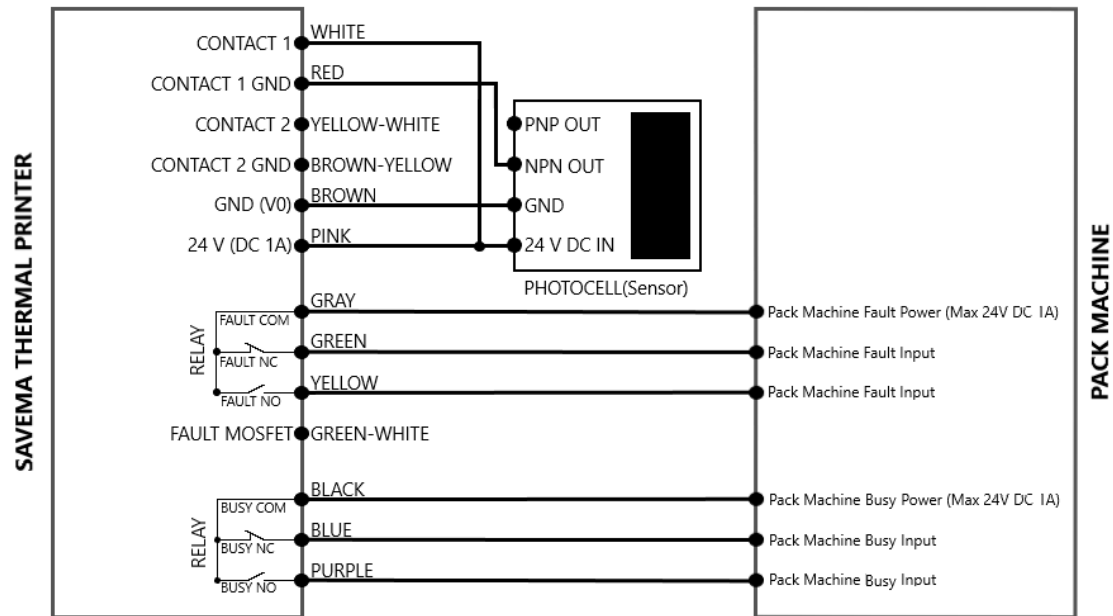


Figure 31. NPN PHOTOCELL Connections

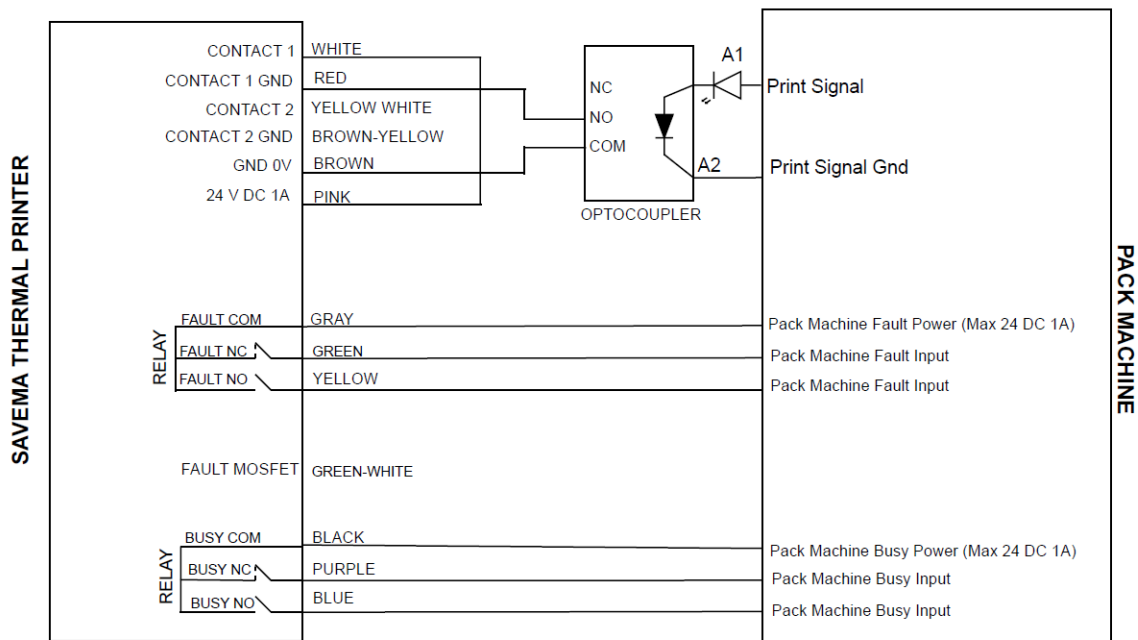


Figure 32 NPN Optocoupler Connections

2.2.3 FAULT MOSFET CONNECTION

Fault mosfet output can be used if fault signal output is wanted to receive the error signal from the mosfet rather than from the relay. 24 V on the printer's contact cable (pink color cable) to the A1+ pin of the external relay. The fault mosfet output on the printer's contact cable (green-white color cable) is also connected to A2- pin of the external relay.

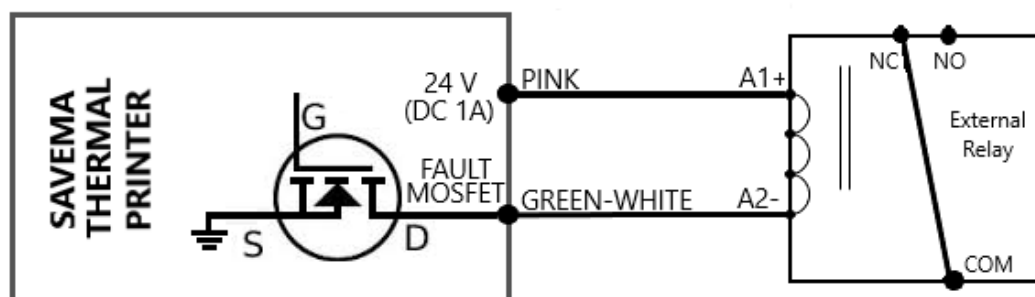


Figure 33. Fault Mosfet Connection

3

INSTALLATION

3.1 ENCODER CABLE

Encoder is the element that gives the machine speed information. Encoder outputs are shown below



Figure 34. Encoder
Spare Part Number: RE



Figure 35. Encoder Cable

No	Designation
1	+24 V
2	GND
3	A+
4	A-
5	B-
6	B+

Table 23. Encoder Cable Pin-Outs

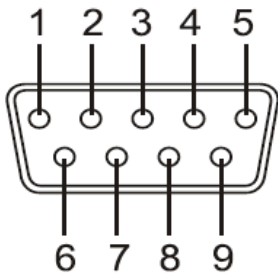
The cable of encoder is connected by an additional connecting cable (a secondary cable) that the other end of which is DB9 male connector.



Figure 36. Encoder Secondary Cable
Spare Part Number: RE-DB9

No	Designation
1	ENCODER SECONDARY CABLE TO ENCODER
2	ENCODER SECONDARY CABLE TO PRINTER

Table 24. Encoder Secondary Cable Descriptions



No	Designation
1/5/6	NO CONNECTION
9	+24 V
4	GND
7	A+
2	A-
3	B-
8	B+

3.2 POWER CABLE

Figure 37. Encoder Secondary Cable DB9 Terminal Numbers

Table 25. Encoder Secondary Cable DB9 Pin-Outs

Power supply AC power input's voltage can change according to country voltage value (110V or 220V). Make sure that the power supply you have purchased meets country standards.

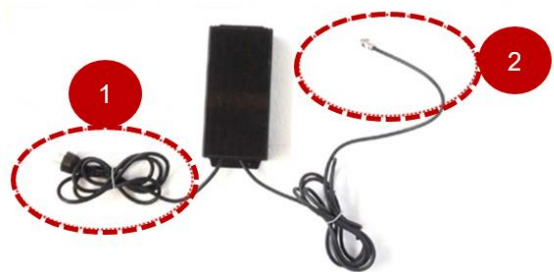


Figure 38. Power Supply
Spare Part Number: GK-32

No	Designation
1	POWER SUPPLY AC INPUT CABLE
2	POWER SUPPLY DC OUTPUT CABLE

Table 26. Power Supply Input and Output

DC voltage pins from power supply are shown below.



Figure 39. Power Supply DC Output Cable Terminal Numbers

No	Designation
1/2	+30 V DC
3	GND
4/5	-V (0 V)

Table 27. Power Supply Output Cable Pin-Outs

The DC output cable of the power supply is connected by a secondary cable that the other end of which is DB15 female connector.

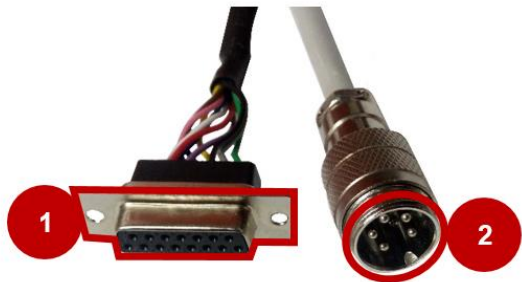


Figure 40. Power Supply Secondary Cable
Spare Part Number: GK-DB15

No	Designation
1	POWER SUPPLY SECONDARY CABLE TO PRINTER
2	POWER SUPPLY SECONDARY CABLE TO POWER SUPPLY

Table 28. Power Supply Secondary Cable Descriptions

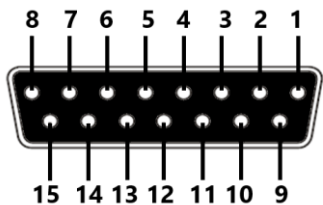


Figure 41. Power Supply Secondary Cable DB15 Terminal Numbers

No	Designation
1/2/10/11	30 V DC
4/5/13/14	-V (0 V)
7/15	GND

Table 29. Power Supply Secondary Cable DB 15 Pin-Outs



Figure 42. Power Supply Cable Connections

No	Designation
1	30 V DC
2	30 V DC
3	-V (0 V)
4	-V (0 V)
5	GND
6	NOTR
7	LINE

Table 30. Power Supply Cable Connections

3.3 PRINTER CABLES



Figure 43. Printer Common Socket

The cables attached to the printer are gathered in a common socket. This common socket contains contact, screen, encoder and power supply sockets.

No	Designation
1	MALE ENCODER SOCKET
2	FEMALE POWER SUPPLY SOCKET
3	MALE CONTACT SOCKET
4	FEMALE SCREEN SOCKET

Table 31. Printer Common Socket Descriptions



Figure 44. Screen Cable
Spare Part Number: C-DATAV7-43

No	Designation
1	SCREEN CABLE TO PRINTER
2	SCREEN CABLE TO SCREEN

Table 32. Screen Cable Descriptions

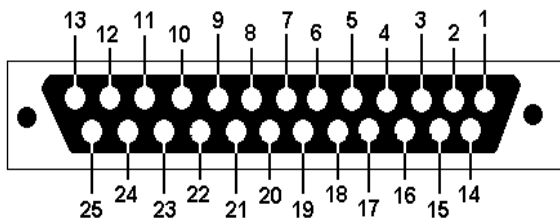


Figure 45. Screen Cable Printer Side Terminal Numbers

No	Designation
24	+30 V
16/17	GND
5	Rx +
18	Rx
6	Tx +
19	Tx
OTHERS	NO CONNECTION

Table 33. Screen Cable Printer Side
Terminal Explanations

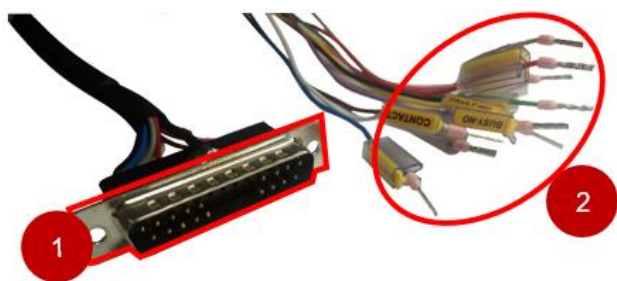


Figure 46. Screen Cable Screen Side Terminal Numbers

No	Designation
1	+30 V
2	GND
3	Rx +
4	Rx
5	Tx
6	Tx +

Table 34. Screen Cable Screen Side Terminal Explanations

One side of the contact cable is attached to the printer, while the other side is left to the user to connect it to the required places. The contact cable contains the cables carrying the busy, fault and contact signals.

Figure 47. Contact Cable
Spare Part Number: C-CNTV7

No	Designation
1	CONTACT CABLE TO PRINTER
2	CONTACT CABLE TO USER

Table 35. Contact Cable Description



The contact cable that belong with “SVM20c Series 32 CK” can be used on “SVM 32CK v6” printer but the contact 2, contact 2 GND and fault-mosfet outputs **must be kept closed** and **must not touch any place**. Otherwise, may cause serious damage to the printer.

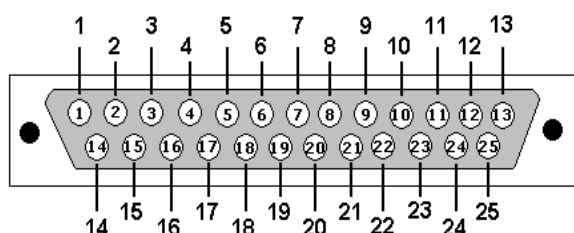


Figure 48. Contact Cable Printer Side Terminal Numbers

No	Designation	Explanation	Color(User Side)
1	BUSY COM	BUSY COM	BLACK
2	BUSY N.C.	BUSY RELAY NORMALLY-CLOSED CONTACT PIN	BLUE
3	FAULT COM	FAULT COM	GRAY
4	-	-	-
5	CONTACT 1 GND	GND 1 PIN OF THE PRINT SIGNAL TO WRITE	RED
6/7/8/9	-	-	-

10	CONTACT 2 GND	GND 2 PIN OF THE PRINT SIGNAL TO WRITE	BROWN-YELLOW
11/12/13	-	-	-
14	BUSY N.O.	BUSY RELAY NORMALLY-OPEN CONTACT PIN	PURPLE
15	FAULT N.O.	FAULT RELAY NORMALLY-OPEN CONTACT PIN	YELLOW
16	FAULT N.C.	FAULT RELAY NORMALLY-CLOSED CONTACT PIN	GREEN
17	CONTACT 1	CONTACT 1 PIN OF THE PRINT SIGNAL TO WRITE	WHITE
18/19/20/21	-	-	-
22	CONTACT 2	CONTACT 2 PIN OF THE PRINT SIGNAL TO WRITE	YELLOW-WHITE
23	FAULT MOSFET	FAULT MOSFET CONTACT PIN	GREEN-WHITE
24	LOW-GND	-	BROWN
25	LOW-24 V	24V DC (MAX 1A) OUTPUT	PINK

Table 36. Contact Cable Printer Side Terminal Explanations

BUSY COM: This pin must be connected to the busy power output of the packaging machine. Busy power output of the packaging machine must be maximum 24 volt DC and 1 ampere. This connection must be done to use the busy signal.

BUSY N.C.: This pin is active when the printer does not produce a busy signal. If it is active, it transmits the voltage that coming from the busy com to the output.

BUSY N.O.: This pin is active when the printer produces a busy signal. If it is active, it transmits the voltage that coming from the busy com to the output.

Busy Com, Busy N.C. and Busy N.O. connections are illustrated in section 2.

FAULT COM: This pin must be connected to the fault power output of the packaging machine. Fault power output of the packaging machine must be maximum 24 volt DC and 1 ampere. This connection must be done to use the fault signal.

FAULT N.C.: This pin is active when the printer does not produce a fault signal. If it is active, it transmits the voltage that coming from the fault com to the output.

FAULT N.O.: This pin is active when the printer produces a fault signal. If it is active, it transmits the voltage that coming from the fault com to the output.

FAULT MOSFET: This pin gives a fault signal output when printer produce a fault signal even if the fault relay fails.

Fault Com, Fault N.C., Fault N.O. and Fault Mosfet connections are illustrated in section 2.

CONTACT 1: The contact signal is required for the printer to print. If this signal is to be received from outside, the + pin of the signal must be connected to this pin.

CONTACT 1 GND: - pin of the signal that + pin is connected to contact 1 is connected to this pin. Contact 1 must be selected from the parameters in order to use contact 1. External print signal setting/External print signal source parameter should be entered 0. (For more detail, see section 1.1.5.1). Contact 1 connections are mentioned in section 2.2.

CONTACT 2: Contact 2 is designed for use in the event of a contact 1 input failure. The + pin of the print signal must be connected to this pin. Contact 2 and contact 1 cannot be active at the same time. In order to use contact 2, External print signal setting/External print signal source parameter should be entered 1.

CONTACT 2 GND: - pin of the signal that + pin is connected to contact 2 is connected to this pin. Contact 2 connections are same as contact 1 connections mentioned in section 2.2. That is, the connection to Contact 1 is made to Contact 2 and the connection to Contact 1 GND is made to Contact 2 GND.

LOW-24 V: It is the + pin of the supply power to the system to be used to generate the contact signal.

LOW-GND: It is the - pin of the supply power to the system to be used to generate the contact signal. (See section 2.2 for Low-24 V and Low-GND connection diagram)

3.4 MACHINE I/O CONNECTION DIAGRAM

The cables attached to the machine's and screen's casing are shown below;

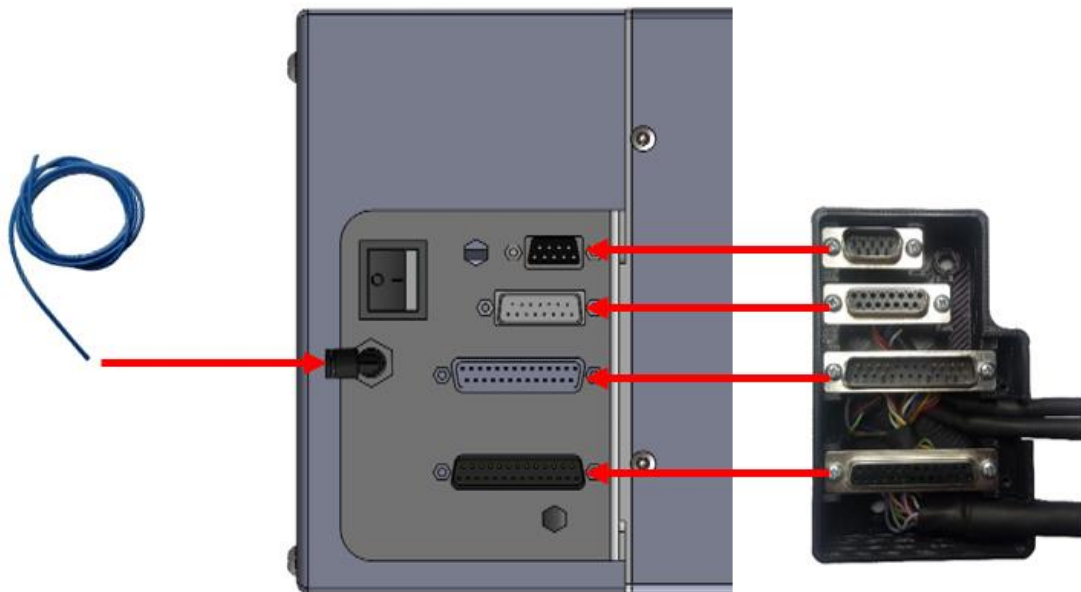


Figure 49. Contact, Encoder, Screen, Power Cable and Air Hose Sockets from Printer Side



Figure 50. Screen Cable Socket from Screen Side

3.5 THERMAL PRINTER ELECTRONIC INSTALLATION

At the time of mechanical installation, make sure there is no power in the system when disconnecting cables. Remove all cables when it is clear that there is no power in the system.

3.5.1 TPH INSTALLATION

Control to thermal print head position is middle of the rubber roller. And be sure the distance between print head and rubber roller is maximum 1.5mm.

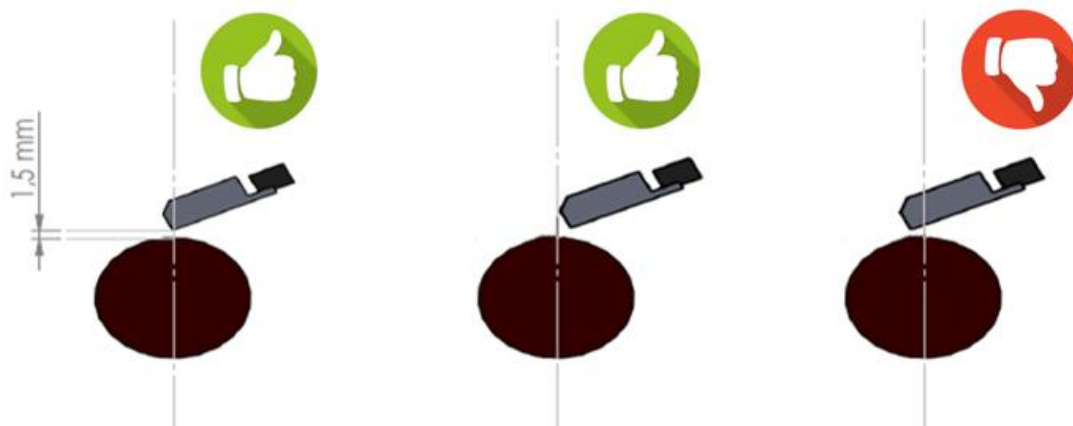


Figure 51. TPH and Rubber Roller Adjustment

The printing part of the thermal head must be parallel to the surface area being printed.

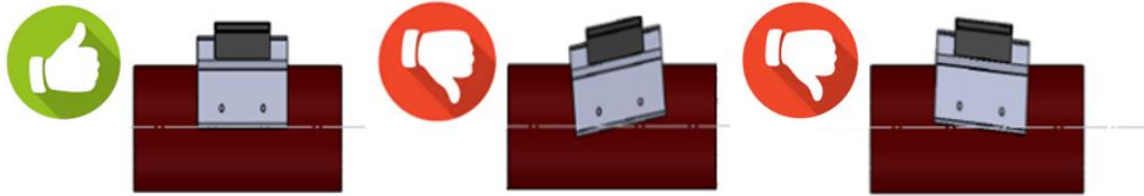


Figure 52. Top View of Thermal Print Head

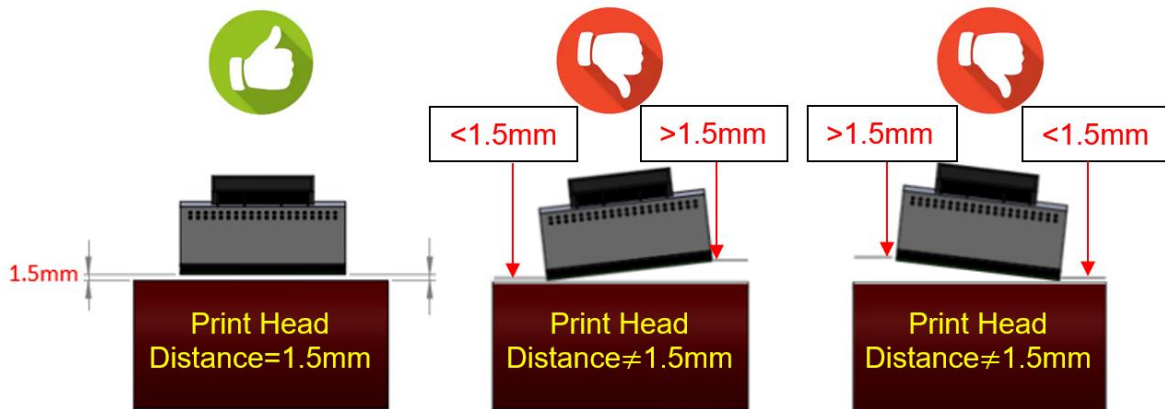


Figure 53. Rear View of the Thermal Print Head

3.5.1.1 THERMAL PRINTER HEAD PRINT POSITION

Continuous printer prints while pack is running. If pack stop while printer is printing, printer cancel printing or not print. Printer Print time and pack machine Stop time must be different time. Continuous printer needs setting for this time conflict. If occur like this problem print position must set from adjustment roller or bracket installation or bracket pipe. When pack machine stop thermal print head position must not on print place.

3.5.1.1.1 Thermal Print Head with **CORRECT** Pack Stop Positions

When the package machine stopped, package must be set up as below for printer.

Solution 1: Print area should be before thermal print head.

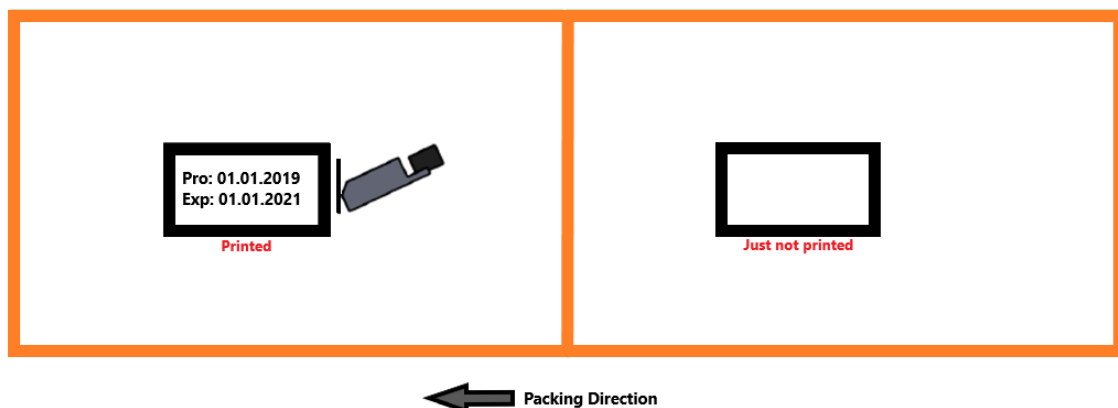


Figure 54. Correct TPH Stop Position Solution 1

Solution 2: Thermal print head must be minimum 15 mm before print place.

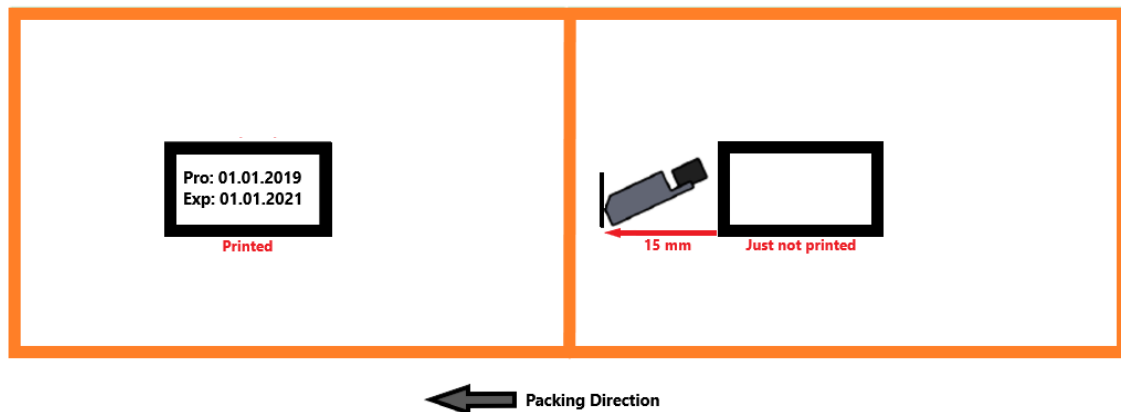


Figure 55. Correct TPH Stop Position Solution 2

Solution 3: Thermal print place must be middle of two print places.

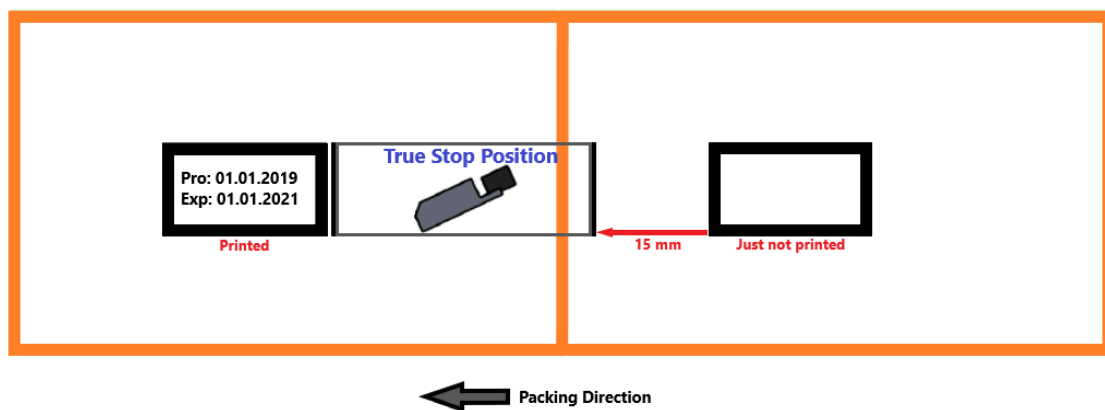


Figure 56. Correct TPH Stop Position Solution 3

3.5.1.1.2 Thermal Print Head with **WRONG** Pack Stop Position

When the package machine stopped, if thermal print head is like below printer does not print.

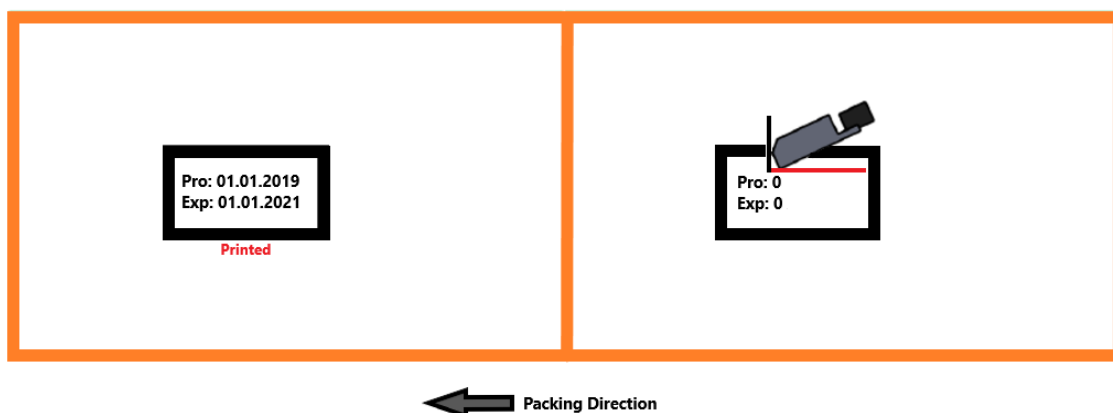


Figure 57. False TPH Stop Position

3.5.2 ENCODER INSTALLATION

Install the encoder like red box in technical picture. Encoder settings/encoder rotation direction parameter must be change "1 or 0" for encoder turn way.

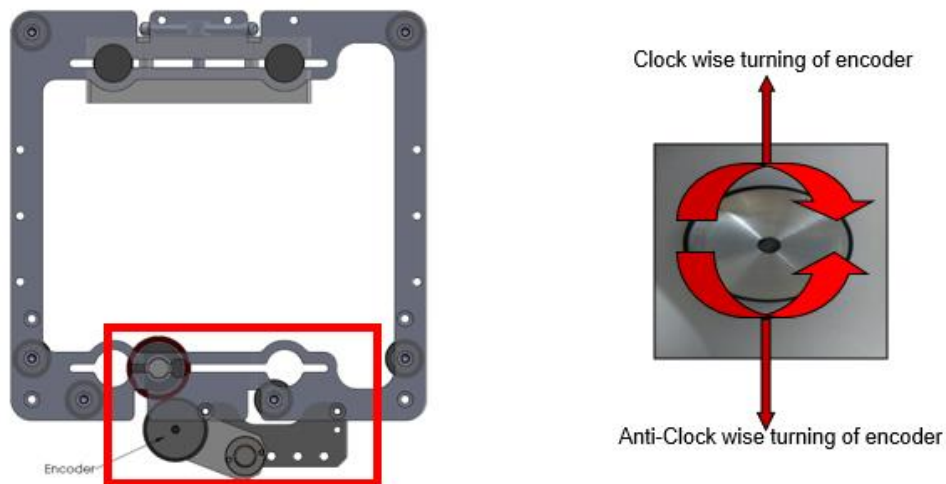


Figure 58. Encoder Position and Turning Directions

After completing all installations, connect all cables and energize the system. Make sure plug on the all cable well.

4

TECHNICAL INFORMATION

4.1 BOARDS AND CONNECTIONS

The correct positions of the boards and the correct connections of the cables are shown below;

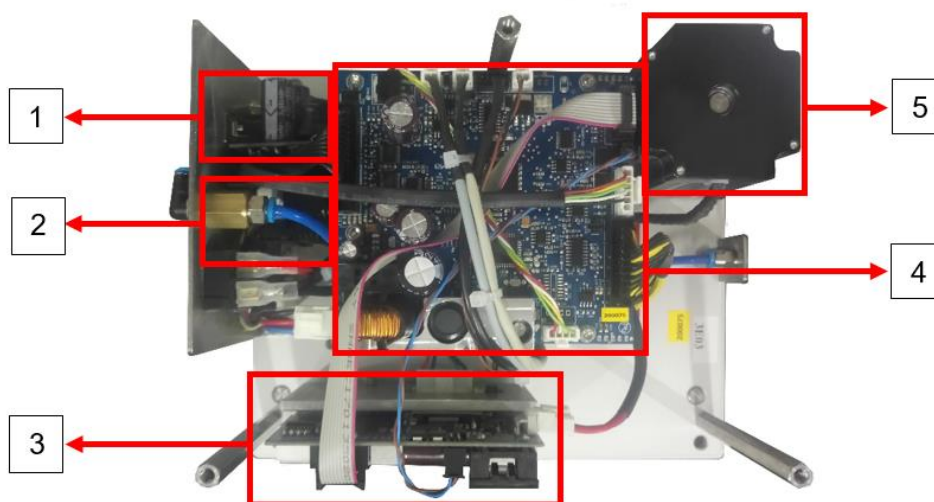


Figure 59. Printer Electronic Boards Positions

No	Designation	Spare Part Number
1	SCREEN COMMUNICATION BOARD(POWER BOARD)	200007-YI
2	CONTACT BOARD	200007-YO
3	MOTOR DRIVER	200007-MDB/C
4	MAINBOARD	200007-MB/C
5	RIBBON MOTOR	200007-M9

Table 37. Printer Electronic Boards Descriptions

4.1.1 MAINBOARD

The names and locations of the sockets on the mainboard are shown below;

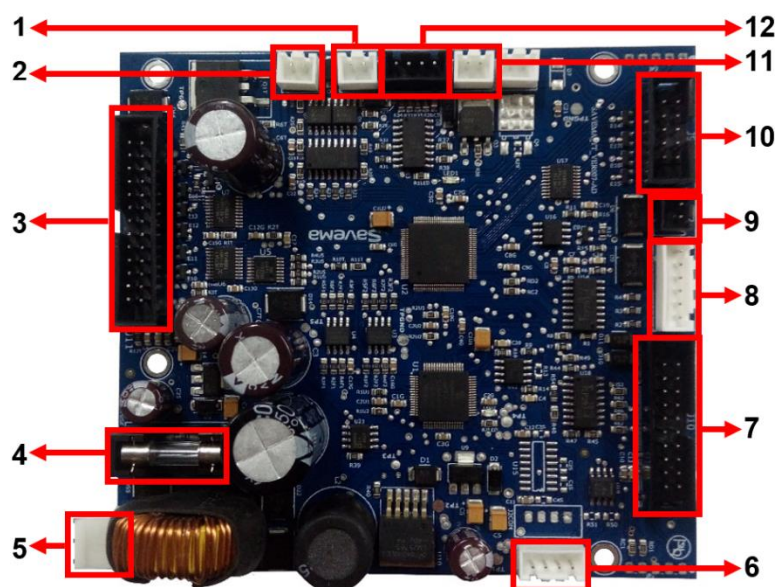


Figure 60. Connectors in Mainboard

No	Designation
1	STOP VALVE SOCKET
2	PRINT VALVE SOCKET
3	PRINthead CABLE SOCKET
4	FUSE 2A
5	POWER INPUT SOCKET
6	SCREEN COMMUNICATION SOCKET
7	CONTACT BOARD SOCKET
8	CASSETTE AND RIBBON SENSORS SOCKET
9	MOTOR DRIVER CANBUS SOCKET
10	MOTOR DRIVER COMMUNICATION CABLE SOCKET
11	PRINthead HEATER SOCKET
12	SWITCH SOCKET

Table 38. Connector Descriptions in Mainboard

4.1.1.1 MAINBOARD CHANGING

Mainboard can be changed following the steps below.

1. Disconnect power connection.
2. Take off all cable in mainboard
3. Remove the screws on the board with Allen wrench.
4. Insert a new board.
5. Replace the screws to removed place and insert all cables.

4.1.2 MOTOR DRIVER BOARD

The names and locations of the sockets on the motor driver board are shown below;

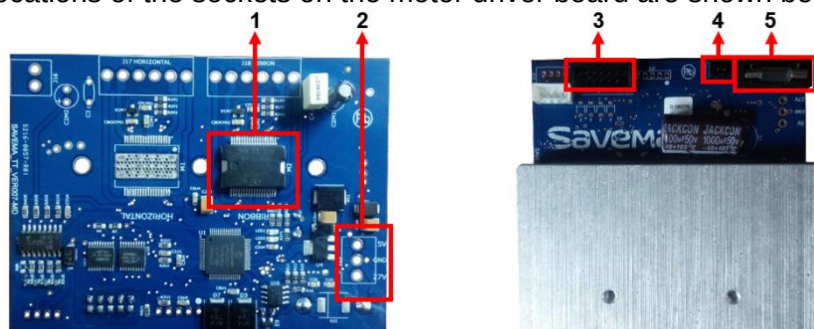


Figure 61. Connectors in Motor Driver Board (Left→Back Side, Right→ Front Side)

No	Designation	Spare Part Number
1	RIBBON MOTOR DRIVER	-
2	POWER INPUT SOCKET	200007-PDP (for cable)
3	MAINBOARD COMMUNICATION CABLE SOCKET	200007-MDC (for cable)
4	CANBUS SOCKET	200007-MDCB (for cable)
5	FUSE 3A	200007-F3

Table 39. Connector Descriptions in Motor Driver Board

4.1.2.1 MOTOR DRIVER BOARD CHANGING

Motor driver board can be changed following the steps below.

1. Disconnect power connection.
2. Take off all cable in motor driver board
3. Remove the screws on the board with Allen wrench.
4. Insert a new board.
5. Replace the screws to removed place and insert all cables.

4.1.3 SCREEN COMMUNICATION BOARD (POWER BOARD)

The names and locations of the sockets on the screen communication board are shown below;

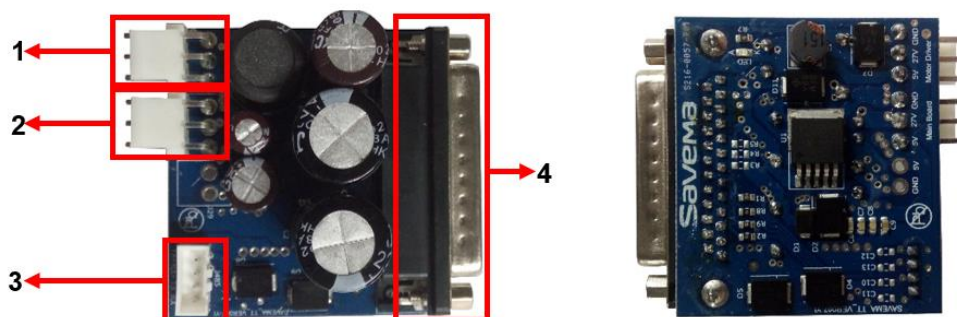


Figure 62. Connectors in Screen Communication Board (Left→Front Side, Right→ Back Side)

No	Designation	Spare Part Number
1	MOTOR DRIVER POWER OUTPUT SOCKET	200007-PDP (for cable)
2	MAINBOARD POWER OUTPUT SOCKET	200007-PMP (for cable)
3	UART CABLE SOCKET	200007-MPC (for cable)
4	DB25 MALE SCREEN COMMUNICATION CABLE CONTACTOR	-

Table 40. Connector Descriptions in Screen Communication Board

4.1.3.1 SCREEN COMMUNICATION BOARD CHANGING

Screen communication board can be changed following the steps below.

1. Disconnect power connection.
2. Take off screen connector cable from machine case (from screen connector).
3. Take off all cable in screen communication board.
4. Remove the screws with Allen wrench (from screen connector).
5. Insert a new board.
6. Replace the screws to removed place and insert all cables.

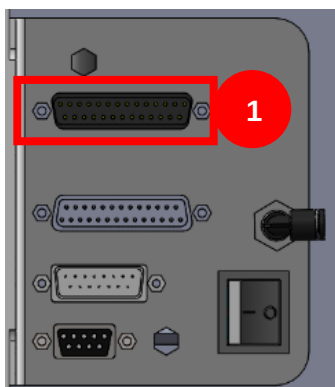


Figure 63. Printer Sockets

No	Designation
1	SCREEN CONNECTOR / DB25 MALE

Table 41. Socket Description

4.1.4 CONTACT BOARD

The names and locations of the sockets on the contact board are shown below;

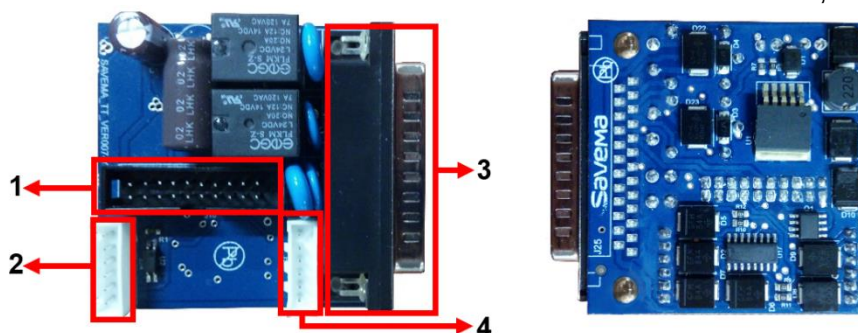


Figure 64. Connectors in Contact Board (Left→Front Side, Right→ Back Side)

No	Designation	Spare Part Number
1	MAINBOARD CABLE SOCKET	200007-MCC
2	COVER LEDS CONNECTION	200007-LD (for LED Board)
3	DB25 FEMALE CONTACT CABLE CONNECTOR	-
4	ENCODER CABLE SOCKET	-

Table 42. Connector Descriptions in Contact Board

4.1.4.1 CONTACT BOARD CHANGING

Contact board can be changed following the steps below.

1. Disconnect power connection.
2. Take off I/O connector cable from machine case (from I/O connector).
3. Take off all cable in contact board
4. Remove the screws with Allen wrench (from I/O connector).
5. Insert a new board.
6. Replace the screws to removed place and insert all cables.

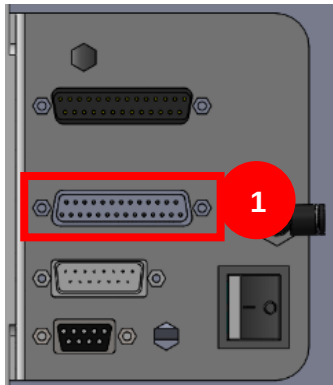


Figure 65. Printer Sockets

No	Designation
1	I/O CONNECTOR / DB25 FEMALE

Table 43. Socket Description

4.2 PACKAGE WINDING SYSTEM

Continuous printers get speed automatically from encoder. Encoder must turn with pack properly. If encoder does not turn with pack properly it possible below problems:

- 1.) Ribbon Broken
- 2.) Missing Print out
- 3.) Bad Ribbon space
- 4.) Each template different size

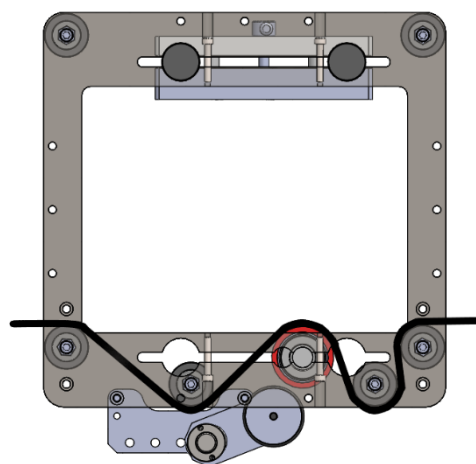
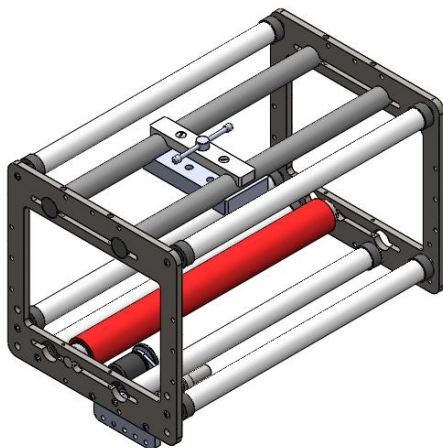


Figure 66. Correct Package Connections

4.3 RIBBON BREAK SYSTEM

The ribbon and cassette sensors in the machine are shown below;

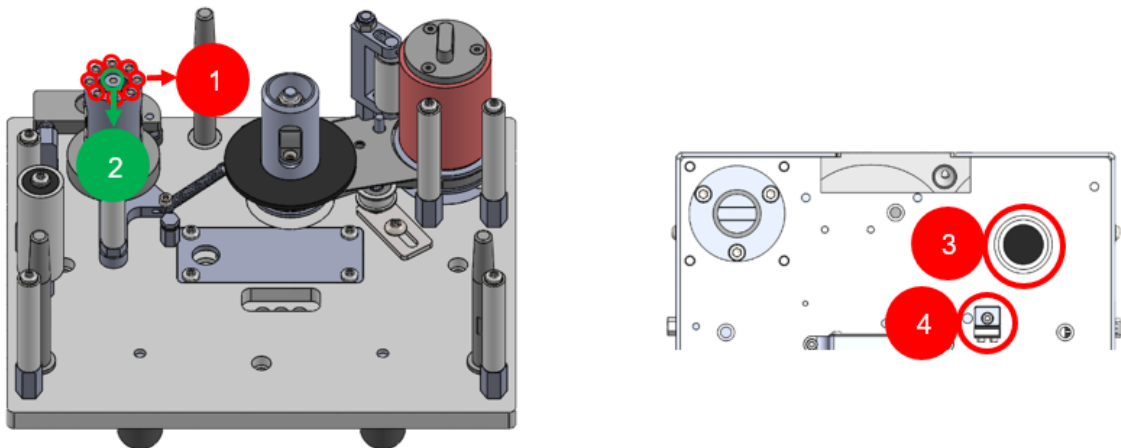


Figure 67. Cassette (Left) and Cassette Slot (Right)

No	Designation	Spare Part Number
1	RIBBON SENSOR TRANSMITTER	-
2	CASSETTE SENSOR TRANSMITTER	-
3	RIBBON AND CASSETTE SENSOR RECEIVER	200007-CS
4	RIBBON BREAK SWITCH	200007-SWU

Table 44. Sensor Descriptions

4.4 THERMAL PRINT HEAD AND PRINTING SURFACE CLEANING

Important warning: Please turn off the power of the printer before cleaning the head. After turning off the power,the head is cleaned with the help of alcohol cotton pad. It takes a while for alcohol to disappear. Then the printer can be used again.

Thermal print head can be cleaned with pure alcohol cotton like picture.

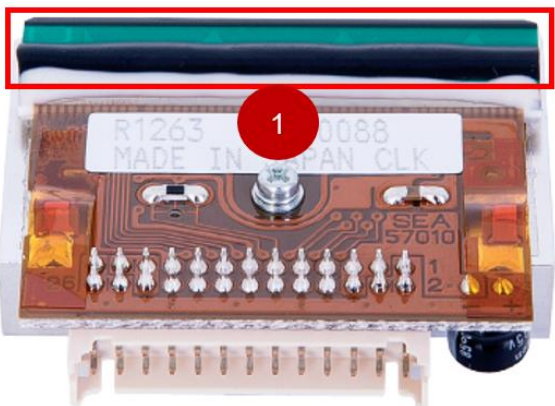


Figure 68. Thermal Print Head
Spare Part Number: 200007-TPH

No	Designation
1	CLEANING AREA

Table 45. Cleaning Area Description



Figure 69. Cleaning of TPH



Figure 70 Printing Surface

Manufacturer's recommendation: If you clean the printing surface and head daily, you will extend the life of the machine.

4.5 RECOMMENDED SPARE PARTS

No	Part Number	Recommended Spare Part Name	Compatibility for SVM 32 CK Version 6 ¹
1	PH	PHOTOCELL	J
2	GK-32	POWER SUPPLY	X
3	200007-M9	STEP MOTOR	X
4	200007-YI	POWER BOARD	X
5	200007-YO	CONTACT BOARD	X
6	200007-MB/C	MAINBOARD	X
7	200007-MDB/C	MOTOR DRIVER BOARD	X
8	200007-TCCB	TPH CABLE CONNECTION BOARD	J
9	200007-TC24	TPH CABLE – 24 CM	J
10	200007-TH2	PRINT HEAD HEATER	J
11	200007-SWU	MICRO SWITCH	J
12	200007-NFB	BLACK ON/OFF BUTTON	X
13	200007-F3	FUSE 3 AMPER	J
14	200007-F2	FUSE 2 AMPER	J
15	200007-TPH	THERMAL PRINT HEAD	J
16	200007-CS	CASSETTE SENSOR	J
17	RE	ENCODER	J
18	EM430-10	COM3 BOARD	X
19	C-DATAV7-43	CONNECTION CABLE FOR 4.3 INCH CONTROLLER	X
20	C-CNTV7	CONTACT CABLE FOR ALL PRINTERS	J
21	C-SETV7-43	CONNECTION CABLE FOR 4.3 INCH CONTROLLER + CONTACT CABLE	X
22	RE-DB9	SECONDARY CABLE BETWEEN ENCODER AND PRINTER	X
23	GK-DB15	SECONDARY CABLE BETWEEN POWER SUPPLY AND PRINTER	X
24	200007-MPC	COMMUNICATION CABLE BETWEEN MAINBOARD AND POWER BOARD	J
25	200007-MCC	COMMUNICATION CABLE BETWEEN MAINBOARD AND CONTACT BOARD	J
26	200007-MDC	COMMUNICATION CABLE BETWEEN MAINBOARD AND MOTOR DRIVER BOARD	J
27	200007-MDCB	CANBUS CABLE BETWEEN MAINBOARD AND MOTOR DRIVER BOARD	J
28	200007-PMP	POWER CABLE BETWEEN MAINBOARD AND POWER BOARD	J
29	200007-PDP	POWER CABLE BETWEEN MOTOR DRIVER BOARD AND POWER BOARD	J

Table 46. Recommended Spare Part List

NOTE: Differences between service manual and purchased device may be due to version update. In this case, please contact the manufacturer.

¹ If the table is filled with a “J”, this part can be used on the specified machine. If it is filled with an “X”, using it on the specified machine may damage the machine. Our company is not responsible for this damage.